

Lame Duck by Primary: Effects of Electoral Incentives on U.S. House Representatives

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Abstract

How do electoral incentives motivate representatives to exert effort and shape policy positions? While reelection is widely recognized as representatives' primary motivation in democratic politics, accurately estimating its effects is challenging because representatives self-select into rerunning or retiring decisions. Leveraging legislators' lame duck status after primary elections, this article estimates how electoral incentives affect U.S. House Representatives. Compiling data from 1955 to 2022, I compare legislators who win primaries and remain motivated for reelection to those who lose and exit the race. Through regression discontinuity and difference-in-differences designs, I find that primary losers significantly reduce effort-related activities, as evidenced by decreased roll-call participation, bill sponsorship, and voter communication. In contrast, the results show that legislators do not adjust policy positions for reelection; both winners and losers maintain consistent ideological and party-line adherence. These findings illuminate the scope of electoral incentives' effects and contribute to debates on electoral accountability and democratic representation.

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1 Introduction

Does the pursuit of reelection prompt representatives to work harder or shift their policy stances? Numerous scholars have emphasized the critical role of electoral incentives in democratic systems, arguing that the desire to secure reelection motivates officeholders to remain closely aligned with their constituents' interests and expectations. Elected officials may be motivated by goals other than reelection such as advancing their preferred policy agendas or heightening their standing within political parties, but the prevailing view holds that “[the electoral goal] must be the *proximate* goal of everyone, the goal that must be achieved repeatedly if other ends are to be entertained” (Mayhew, 2004). In this way, a drive for reelection serves as a fundamental mechanism of democratic accountability, ensuring political leaders' responsiveness to voter preferences and setting democratic governance apart from autocratic rule.

Despite the importance of electoral incentives in democratic politics, however, existing research offers conflicting theoretical perspectives and mixed empirical findings on what specific behaviors these incentives elicit from representatives. In theory, electoral incentives could pull a representative's behavior in opposing directions, or fail to move it at all when other forces dominate. Representatives may strive to meet voters' demands to secure reelection (Mayhew, 2004; Fenno, 1978). At the same time, they may (or may not) feel pressure from party leaders to act otherwise as they control crucial campaign and career resources (Rohde, 2010; Aldrich, 2008; Cox and McCubbins, 2005; Krehbiel, 1993). Alternatively, if representatives hold strong personal ideological commitments, electoral incentives may induce little change in their behavior at all (Alesina, 1988; Osborne and Slivinski, 1996; Besley and Coate, 1997; Poole and Rosenthal, 1985).

A core challenge in empirically evaluating theoretical claims about the impact of electoral incentives lies in disentangling these incentives from legislators' strategic decisions about whether to seek reelection or retire. Analyses on electoral accountability often rely on comparison of the behavior of incumbents who run for another term with the behavior

of those who step down. However, rerunning or retiring decisions are shaped by a range of personal and political factors, making it difficult to attribute observed behavioral differences solely to the presence or absence of reelection motives. As a result, much of the literature has relied on theoretical assumptions and correlational evidence. While previous scholars have employed various strategies to address this selection bias, robust empirical evidence remains limited, particularly for legislators in highly salient political positions such as U.S. House Representatives.

This article addresses this empirical challenge in the context of the U.S. House of Representatives by exploiting the unique opportunity presented by incumbent legislators' lame duck status after primary elections. By collecting and combining data from over 3,800 incumbent legislators and their primary election contests between 1955 and 2022, I compare the post-primary behavior of primary election winners, who remain motivated for the upcoming general election, with that of primary election losers, who are disqualified from running for reelection. To ensure comparability between winners and losers, I employ two complementary research designs. A regression discontinuity design compares incumbents who narrowly win or lose primary elections, and a difference-in-differences design contrasts legislators' behavior across terms in which they lost or won renomination.

The findings from both designs provide strong evidence that electoral incentives play a significant role in determining the level of effort legislators invest. After losing primary elections, legislators show a notable decline in roll-call participation, bill sponsorship, cosponsorship, and the frequency of e-newsletters sent to constituents. To further investigate whether it is the pressure of upcoming general elections that drives this behavior, I analyze how these effects vary with the expected competitiveness of each district's general election. I find that the effects are substantial mostly in districts where general elections are expected to be competitive. This suggests that electoral concerns, rather than non-electoral considerations such as the anticipation of serving another term or future committee assignments, are likely the primary drivers of the results.

On the other hand, the findings from both designs show that electoral incentives do not meaningfully affect legislators’ policy positions. Both primary election winners and losers remain largely consistent in terms of ideological extremity and party-line adherence following primary elections, contrary to a long line of theory in which the positions legislators take respond to electoral pressure (e.g., Downs, 1965; Mayhew, 2004). To investigate the extent of this stability, I further restrict the analysis to bills that receive greater attention from party leaders, bills that scholars have marked as significant, and contested bills that are close to passing or failing. The results show that neither primary election winners nor losers significantly change their policy positions even for these bills, underscoring the substantive strength of the null findings.

These findings carry several implications. By distinguishing the activities that respond to electoral incentives from those that do not, they sharpen the foundational assumptions of electoral accountability: the effort results reinforce Mayhew’s premise that reelection is the proximate motivation (Mayhew, 2004), while the policy-position null cuts against the corollary that legislators’ roll-call positions are themselves electorally driven—and, more broadly, against the Downsian account of electoral competition (Downs, 1965). The null instead fits models in which legislators’ ideological stances are largely fixed (Alesina, 1988; Osborne and Slivinski, 1996). The stability of incumbent positions also bears on debates over party influence (Cox and McCubbins, 2007; Krehbiel, 1993), the sources of polarization, and the reliability of roll-call-based ideology measures: it suggests that polarization operates more through electoral replacement of legislators than within-term repositioning (Barber and McCarty, 2015; McCarty, 2019; Moskowitz, Rogowski and Snyder, 2024), and that scores such as DW-NOMINATE (Poole and Rosenthal, 1985; McCarty, 2016) capture stable preferences rather than artifacts of immediate electoral pressure.

1.1 Theoretical Considerations on Legislators' Behavior

To understand how electoral incentives affect legislators' behavior, this article focuses on two key aspects: effort-related behavior and policy positions based on roll-call votes. Standard models of electoral accountability suggest that electoral incentives drive elected officials to increase their effort levels in hopes of securing reelection (Barro, 1973; Ferejohn, 1986). However, when representatives must allocate limited resources across multiple tasks, it is less clear which specific activities will be prioritized (Fenno, 1973; Holmstrom and Milgrom, 1991). For example, electoral incentives may push legislators to attend roll-call votes to avoid perceptions of shirking, while also pulling them away from such floor activity toward campaigning and district demands.

When it comes to legislators' policy positions, theoretical predictions diverge. Some models predict electoral incentives pull candidates toward the median voter (Downs, 1965; Calvert, 1985), while others suggest legislators shift toward positions that mobilize core supporters (Adams and Merrill, 2003; Fiorina, 1974).

A different theoretical tradition, or citizen-candidate and commitment-based models, instead assumes that legislators' policy positions remain largely fixed: candidates are treated as ideologues who steadfastly adhere to their preferred ideological stances (Alesina, 1988; Osborne and Slivinski, 1996; Besley and Coate, 1997). According to this strand of models, electoral incentives would have no effect on legislators' policy positions.

It also remains unclear how much parties influence their legislators through electoral incentives, given the estimation challenge of legislators' selection into parties (Smith, 2007; Krehbiel, 1993; Cox and McCubbins, 2007). If party influence exists, legislators may adopt party-line stances when seeking reelection, since the party supplies campaign resources. Donors and interest groups may exert similar pressure through the same channel, though evidence of their influence is mixed (Stratmann, 2002; Ansolabehere, de Figueiredo and Snyder, 2003; Fowler, Garro and Spenkuch, 2020). Because these external pressures may pull in different directions, the net effect of electoral incentives on policy positions is unclear.

Comparing rerunning and leaving legislators is the most natural empirical strategy, as the former faces the prospect of an upcoming election while the latter does not. However, even setting aside selection differences between the two groups, its interpretation hinges on the motivations and incentives that drive not only the former but also the latter. Leaving legislators are often expected to reduce their legislative activity or coast through their remaining term, but alternative motivations can keep them active in office. For instance, they may anticipate post-Congress appointments to federal or local office, or prepare for careers in the private sector or academia. Personal policy preferences or ideological commitments may also keep them active; freed from electoral pressures, they may pursue these commitments more openly than before. In particular, if leaving legislators themselves remain motivated by future electoral concerns—say, sustaining a record for a later bid—the comparison reflects only a difference in horizon, not the presence or absence of electoral incentives, and cannot identify their effect.

These alternative motivations complicate the interpretation of this comparison as a clean test of electoral incentives. The major complication depends on what causes leaving. Voluntary retirements and anticipated departures by term limits let alternative motivations work through anticipatory behavior, making it difficult to isolate the effects of electoral incentives. A less anticipated event like a primary defeat closes this anticipatory channel, but the comparison’s cleanness still depends on what motivates the unanticipated leavers themselves. Reading the rerunning-versus-leaving comparison as a clean test of electoral incentives therefore requires a closer look at leaving legislators’ motivations and trajectories, which the next subsection takes up as part of its review of previous research.

1.2 Previous Research on Electoral Incentives

Estimating the effects of electoral incentives is challenging because it requires a credible counterfactual for what a legislator would do without them. Previous research has approached this challenge through several distinct designs, each constructing the counterfactual in a

different way.

Earlier literature on shirking examined whether legislators behave differently in their last term, when they are not seeking reelection (Lott, 1987; Lott and Davis, 1992; Zupan, 1990). Because leaving legislators do not face electoral incentives, they can serve as a counterfactual, but only if they are otherwise comparable to those seeking reelection. If retirement decisions are strategic, or if retirees differ in quality or ideology from those seeking reelection, simple comparisons would introduce selection bias. Because these studies relied on such comparisons, they could not separate the effect of electoral incentives from systematic differences in who leaves office.

Building on earlier research, scholars have attempted to address this selection bias through various research designs. These approaches include leveraging term limits (Alt, Bueno de Mesquita and Rose, 2011; Fourniaies and Hall, 2022; Besley and Case, 1995), electoral reforms (Motolinia, 2021), and parties' candidate nomination processes (Fiva and Nedregård, 2024).

Similar to this article, Fiva and Nedregård (2024) employ party renomination decisions in Norway to compare legislators' within-term behavior following the renomination outcome. Both studies find that legislators who fail to secure renomination tend to reduce their effort-related activities while largely maintaining their policy positions. Aside from the comparability concerns inherent to party-led renomination decisions in Fiva's setting, the two works are situated in markedly different political contexts. In Norway, party leaders decide which incumbent legislators are renominated, a process characterized by strong party discipline and limited personal campaigning. In contrast, in the United States, incumbents must compete in primary elections that emphasize individual candidacies. Consequently, while Fiva and Nedregård (2024) highlight the influence of party leadership in shaping legislators' behavior, this article centers on how individual legislators pursue reelection under a more candidate-driven electoral system.

Using a term-limit design with state legislators, Fourniaies and Hall (2022) also report

Table 1: Career Trajectory After Losing in Primary Election

Run for Elections Later	Non-elected Public Position	Non-public Career	Retire or No Records
28 (17.50%)	27 (16.88%)	63 (39.38%)	42 (26.25%)

Note: The table shows the distribution of House representatives’ post-Congress career trajectories after losing in primary elections. Only 17.5% of them ended up running for an electoral position again, while the majority pursued non-electoral or non-public roles, retired, or left no trace of their activities after leaving Congress.

similar findings: termed-out legislators reduce productivity yet maintain policy positions. The key difference is electoral salience. U.S. House races attract far more public attention than state-level contests, and past work attributes the policy-position null among state legislators specifically to that low salience (Rogers, 2017, 2023). The finding that the same null obtains in the high-salience U.S. House context is therefore striking, contrary to claims from earlier studies (e.g., Ansolabehere, Snyder and Stewart, 2001; Griffin, 2006; Fiorina, 1973).

Compared to a term-limit framework, this article’s use of primary elections offers a more precise way to differentiate between legislators motivated by electoral considerations and those who are not. As most House Representatives exit electoral politics after a primary defeat (Table 1), comparing primary winners and losers provides a clearer gauge of the impact of electoral incentives on legislative behavior. By contrast, termed-out state legislators may still respond to future political prospects and party pressure: Steen (2006) shows that those hitting state-legislative term limits are more likely to run for Congress than those still eligible to serve, Fourniaies and Hall (2022) find stronger productivity effects in states where term limits prohibit running for the other chamber, and Olson and Rogowski (2020) document continuing party pressure on termed-out legislators. These ongoing incentives confound any attempt to isolate the effect of reelection pressure.

Other studies compare general-election winners and losers in brief November–December postelection lame-duck sessions (Jenkins and Nokken, 2008*a,b*; Schönenberger, 2024), and

they face their own limitations. Such sessions see little new legislative activity since bills introduced then expire with the outgoing Congress, and what business does occur is largely confined to must-pass items—appropriations, the National Defense Authorization Act, and continuing resolutions—shaped by strategic party agenda-setting. Furthermore, general-election winners with a two-year horizon are less immediately motivated by reelection than primary winners, who face a general election within months and provide a sharper baseline.

Finally, Cowburn and Sältzer (2025) employ a research design most similar to this article’s by comparing winners and losers of primary elections among U.S. House representatives. They analyze legislators’ language usage on X (Twitter) in 2020 and find that Democratic primary losers took significantly more moderate positions after their defeat, whereas no such effects were observed among Republicans. By contrast, this article examines a wider array of legislative behavior, focusing on various effort-related activities and policy positions based on roll-call votes, over a more extensive period from 1955 to 2022.

2 Data on Primary Elections and Legislator Behavior

In the United States, primary elections typically occur several months before the November general election—as early as March and as late as September—providing parties an opportunity to select their preferred candidate to compete for the office. For incumbent legislators, these contests are especially significant: a primary win maintains their eligibility and drive to compete in the general election, while a defeat effectively disqualifies them from running for reelection.

This article exploits these features of primary elections by comparing the post-primary behavior of incumbent House representatives who win their primary contests with that of those who lose.

Pettigrew, Owen and Wanless (2014) provide primary election results for U.S. House Representatives from 1956 to 2010, and Miller and Camberg (2020) provide them from 2012

to 2018. These data do not include runoff elections, so I manually collected them from state archives and past newspapers; I also collected results for 2020 and 2022 from state archives. The combined data cover U.S. House Representatives from 84th to 117th Congress, or 1955 to 2022, a total of 14,542 Congress-legislator units, and 2,668 distinct legislators. Of them, I focus on 3,860 Congress-legislators (1,546 distinct legislators) who competed in primary elections with at least one other challenger. There were 3,700 victories and 160 defeats.

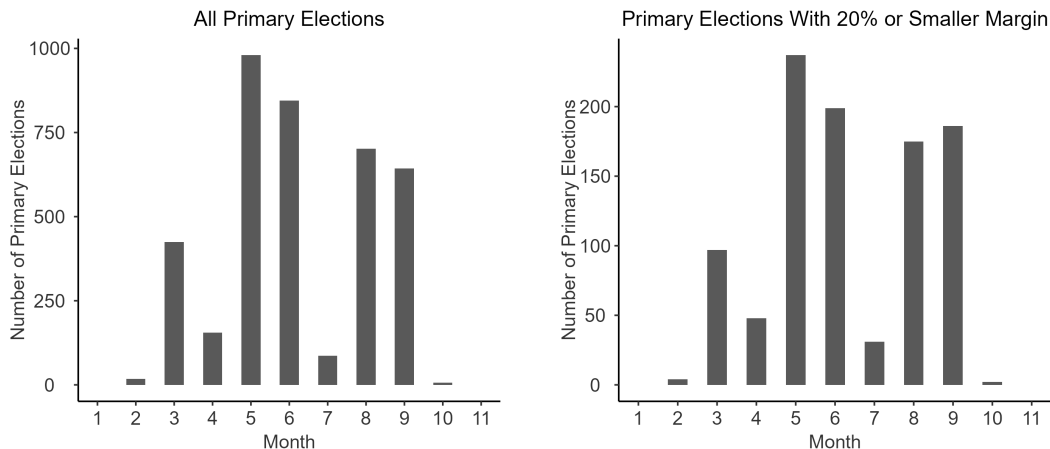
To measure policy positions, I use roll-call votes, including ideological positions on a liberal-conservative scale and party-line stances. For effort-related behavior, I examine legislative activities that proxy for legislator effort, drawing on previous literature (Dal Bo and Rossi, 2011; Titunik, 2016; Fourinaies and Hall, 2022). These activities include roll-call participation, bill sponsorship and cosponsorship, and the frequency of e-newsletters sent to constituents.

The House representatives' information and roll-call voting records are collected from [voteview.com](https://www.voteview.com) (Lewis et al., 2024). Sponsorship data come from the Congressional Bills Project and cover 1973 to 2018 (Adler and Wilkerson, 2020), and James Fowler's cosponsorship network data provide cosponsorship dates from 1979 to 2004 (Fowler, 2006). E-newsletter data come from DCinbox (Cormack, 2024) and cover 2010 to 2022. Because legislators' general election results may affect their electoral incentives, I exclude activities after the general election date from the analyses.

Examining post-primary behavior requires knowing the exact dates of primary elections. The Federal Election Commission provides primary election dates for all states back to 1996, and a few state archives provide earlier dates. For the other years and states back to 1956, I manually collected them from historical newspaper articles.

Figure 1 and Figure 2 show the distribution of primary election dates by month, state, and time. The median month of primary elections was June. A total of about three million roll-call votes were cast by the legislators in the analysis, and 22 percent of those were cast after each legislator's primary election date.

Figure 1: Monthly Distribution of Primary Elections



Note: The left figure shows the month distribution of all primary elections, while the right figure shows those with less than 20% point margins. In both cases, June was the median month.

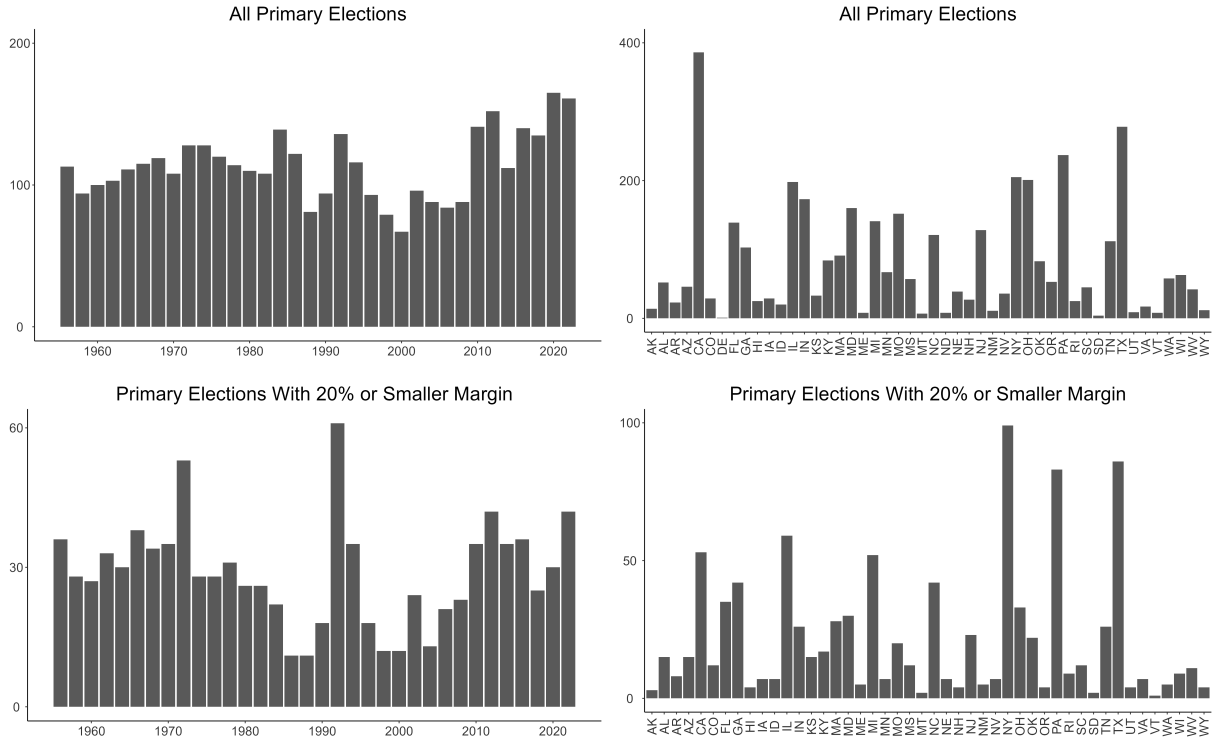
3 Empirical Strategy

To examine the behavior of primary winners and losers, I employ two complementary research designs. A regression discontinuity design compares legislators who narrowly won versus narrowly lost close primary races. A difference-in-differences approach tracks how legislators' behavior changes across terms in which they won primaries versus terms in which they lost. Both designs aim to ensure comparability between winners and losers: the regression discontinuity design exploits the as-if random outcomes of close primary elections, while the difference-in-differences approach uses within-legislator changes across terms. Each design has its own strengths and weaknesses, and together they provide more robust evidence on the effects of electoral incentives.

3.1 Variables and Measurement

The main independent variable distinguishes whether an incumbent House member wins or loses a primary election, capturing the presence or absence of electoral incentives for the general election. The dependent variable tracks how legislators behave after the primary, reflecting their response to the electoral incentives. Specifically, I measure within-term change

Figure 2: Distribution of Primary Elections by Year and State



Note: The graphs on the left and right illustrate the number of primary elections by year and by state, respectively. The top rows display the distribution for all primary elections, and the bottom rows show the distribution for those with margins of less than 20%.

in behavior from pre-primary to post-primary periods, which discounts varying base levels across legislators and absorbs any pre-primary anticipatory behavior into the baseline, thereby isolating each legislator's reaction to primary outcomes.

This article's outcome variables fall into two structurally distinct types. Vote-based outcomes—roll-call participation, party-line voting, and ideological positions—are responses to a legislative agenda set externally by party leaders and committee chairs: legislators respond to whichever bills reach the floor. Because the composition of bills scheduled before and after the primary may differ systematically, controlling for bill-level variation is essential. In contrast, non-vote outcomes—bill sponsorship, cosponsorship, and e-newsletters—are activities where legislators can more freely set their own agenda. This structural difference

shapes the dependent variable construction and estimation strategy for each type.

To analyze non-vote behaviors, I construct a dependent variable that reflects the proportion of these activities conducted after the primary election. For example, if a legislator sponsors six bills before the primary and four after, the dependent variable is calculated as the proportion of post-primary sponsorship, or $\frac{4}{6+4} = 0.4$. Since primary election dates vary by state and year, legislators in states with later primary dates have shorter post-primary windows, so a simple pre/post comparison would be misleading. To address this, I introduce a reference date t^* for each legislator-year, which is set earlier than the primary date so that the time length between t^* and the primary is equal to the time length between the primary and the general election (Figure 3). Pre-primary activity B_1 is measured between t^* and t_2 , and post-primary activity B_2 between t_2 and the general election date t_3 . The change in behavior is then measured by the proportion of post-primary activities, or

$$\frac{B_2}{B_1 + B_2}, \quad (1)$$

whose value higher than 0.5 signifies greater engagement in that behavior after the primary election. I apply this measure to legislators' bill sponsorship, co-sponsorship, and e-newsletter counts. This construction is used in both the regression discontinuity and difference-in-differences designs.

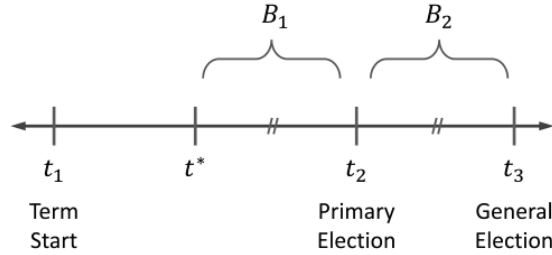
For roll-call-based behaviors (roll-call participation, voting with the party majority, and voting with party leaders), the two research designs construct outcomes at different units of analysis but estimate the same within-legislator post-primary parameter.

For the regression discontinuity design, I construct a legislator-level dependent variable by estimating, for each Congress t , the following OLS regression:

$$\text{Vote}_{ipbt} = \gamma_{it} + \delta_{pb} + \beta_{it}\text{AfterPrimary}_{ibt} + \varepsilon_{ipbt}. \quad (2)$$

In this model, the left hand side, Vote_{ipbt} , is a binary indicator set to 1 if legislator i from

Figure 3: Dependent Variables for Effort-related Behavior



Note: This figure explains how I measure legislators' change in effort-related behavior after primary elections. A reference date t^* is set such that the length of the period between t^* and the primary election date t_2 is the same as that between the primary election date t_2 and the general election date t_3 . B_1 and B_2 measure a legislator's behavior in the period between t^* and t_2 , and t_2 and t_3 , respectively. Then the dependent variable measures a legislator's change in behavior after the primary election by $\frac{B_2}{B_1+B_2}$.

party p exhibits a specific voting behavior, such as participating in roll-call voting, on bill b in Congress t , and 0 otherwise. The term γ_{it} represents legislator fixed effect and δ_{pb} accounts for bill-by-party fixed effects. The key variable, $\text{AfterPrimary}_{ibt}$, is a binary indicator that takes the value 1 if legislator i faces the roll-call for bill b after the primary election and 0 otherwise. The coefficient β_{it} captures the direction and magnitude of the change in legislator i 's voting behavior after the primary in Congress t , and I use it as the legislator-level outcome in the regression discontinuity specification (Section 3.2.1). I extend this approach to other aspects of roll-call voting behavior by re-estimating the regression with alternative definitions of Vote_{ipbt} , such as aligning with the party majority, or voting in line with party leaders.¹

For the difference-in-differences design, I construct a bill-vote-level dependent variable by retaining the same bill-level vote data and estimating the post-primary change directly via the interaction between primary outcome and the post-primary indicator (full specification in Section 3.2.3). This avoids the intermediate step of collapsing each legislator-Congress' voting records into a single β_{it} and estimates the same within-legislator parameter; the two approaches yield quantitatively close estimates (Table A.8).

¹When measuring the policy positions based on roll-call votes, I follow existing measurement methods and include roll-call votes where legislators voted either "yea" or "nay," excluding any abstained votes, to ensure that the policy position estimates are not conflated with abstention behavior.

The two designs construct the vote-based dependent variables at different levels of analysis to reflect the structure of their identifying variation. The regression discontinuity design compares different legislators near the threshold, so each legislator-Congress contributes one independent observation; replicating the same running variable across a legislator’s bill votes would inflate the sample without adding independent comparisons. The difference-in-differences design instead compares the same legislator across terms, so vote-level observations are natural for the within-legislator comparison and allow bill-by-party-by-Congress fixed effects to absorb bill composition differences.

These approaches for vote-based behavior share an important advantage over a simple pre/post mean comparison: the types of bills introduced before and after primary election dates may differ systematically, so a raw rate comparison could conflate genuine post-primary behavior change with shifts in the legislative agenda. Through bill-party fixed effects within Congress, these dependent variables absorb the bill-by-party level differences, ensuring that the estimated change reflects post-primary lame-duck status itself.

Nevertheless, as a robustness check, I also measure policy positions in the appendix using a simple difference of mean voting patterns between pre- and post-primary periods, with consistent results (Table A.7). The appendix also presents a descriptive comparison of bills before and after the primary. The distribution of bill topics is largely stable across the two periods, with government operations, defense, and public lands remaining the most common categories both before and after the primary (Figure A.3); similarly, weekly vote volume, participation, and passage rates show no systematic shift around the primary date (Figure A.4). These comparisons thus reveal no notable, systematic differences in the bills considered before and after primary elections.

Roll-call extremism, an additional vote-based outcome, requires one further step beyond this shared construction: each vote is coded by the legislator’s own-party pole (conservative for Republicans, liberal for Democrats), so that higher values indicate own-pole voting. For the regression discontinuity design, I re-estimate equation (2) with Vote_{ipbt} set to a

conservative-vote indicator—classifying each bill as liberal or conservative using the bill-direction algorithm of Fowler and Hall (2012)—and then sign-flip the resulting β_{it} so that higher values indicate increased conservatism among Republicans or increased liberalism among Democrats. For the difference-in-differences design, the same conservative-vote indicator is sign-flipped for Democrats—yielding the own-party-pole indicator extreme_{iv} —so that higher values likewise indicate own-pole voting. Both approaches estimate the same conceptual quantity, i.e., the within-legislator post-primary shift in own-pole voting, and I additionally examine the absolute values of these changes to assess whether legislators adjust positions in either direction even when net shifts cancel.

3.2 Research Designs

The research designs exploit the fact that an incumbent must win the party’s primary or be a sole candidate to advance to the general election. While most incumbent House Representatives secure primary election victories with relative ease, this is not always the case. Occasionally, a prominent local figure’s entry into politics poses a threat to the incumbent, and at other times, primary voters view their incumbent politician as old and incompetent, opting to replace them with a new candidate. Changes in district boundaries can also jeopardize an incumbent’s electoral prospects, as they may move loyal supporters to other districts. Consequently, some primary elections result in either the incumbent losing or facing a closely contested race. These variations in outcomes provide the identifying variation for estimating electoral incentive effects.

3.2.1 Regression Discontinuity Design

First, I implement the regression discontinuity design. The running variable V_{it} is a legislator i ’s share of the top two candidates’ vote less 0.5 in the primary election during Congress t . For example, if an incumbent received 60% of the votes, and two challengers received 35% and 5%, respectively, the running variable is calculated as $\frac{60\%}{60\%+35\%} - 0.5 = 0.132$. This

indicates that incumbents with positive running variables won primary elections, whereas those with negative running variables lost.

The construction of this running variable is not feasible in states with different primary election systems, so I exclude these states. Specifically, I exclude Georgia before 1962, which used a county-unit system in place of direct popular vote. I also exclude states with nonpartisan blanket primaries, such as the top-two or top-four systems used in Washington (since 2008), California (since 2012), and Alaska (since 2022). Additionally, I exclude Connecticut and Louisiana because their primary systems differ significantly; in Connecticut, party nominees are typically chosen at party conventions, making primaries rarely competitive, while Louisiana uses a nonpartisan blanket primary where winners can be elected outright.

I estimate the standard regression discontinuity specification,

$$Y_{it} = \beta_0 + \beta_1 \mathbf{1}\{V_{it} > 0\} + f(V_{it}) + \varepsilon_{it}. \quad (3)$$

Here, V_{it} is the running variable, or a legislator’s top-two winning margin in the primary election, and Y_{it} captures the legislator’s change in behavior following the primary election. The function $f(\cdot)$ is flexible in the running variable; given the small sample size, I primarily use a linear specification (Gelman and Imbens, 2019) with a 20-percentage-point bandwidth. In the appendix, I show that the sign and direction of the estimates remain largely stable across alternative fixed bandwidths (Figure A.2), and I report estimates at the data-driven optimal bandwidth of Calonico, Cattaneo and Titiunik (2014) (Table A.2). Because close primaries are few, those optimal windows are narrow and trade statistical power for local identification—several effort coefficients, including participation, lose significance there—so I treat them as a limitation of the close-primary design rather than as corroboration and retain the 20-percentage-point window for the main analysis. The coefficient of interest, β_1 , represents the RD estimator, capturing the effect of primary election victory on changes in legislators’ behavior.

Table 2: Observable Characteristics of Legislators by Electoral Margin in Primaries

	(1) Close Losers	(2) Close Winners	(3) Close Primaries	(4) Non-close Primaries
N	159	820	979	2,881
Congress	100.08 (10.31)	100.16 (10.39)	100.14 (10.37)	101.34 (10.04)
Seniority	5.11 (3.31)	5.04 (4.01)	5.05 (3.90)	5.06 (3.90)
Age	57.99 (10.98)	56.27 (10.97)	56.55 (10.98)	55.40 (11.01)
Democratic party	0.65	0.62	0.63	0.62
Previous vote share in general	0.68 (0.15)	0.71 (0.16)	0.70 (0.15)	0.70 (0.15)
DW-NOMINATE score	-0.08 (0.40)	-0.05 (0.39)	-0.05 (0.40)	-0.06 (0.40)
DW-NOMINATE $\times (-\mathbf{1}\{\text{Democrat}\})$	0.37 (0.18)	0.35 (0.19)	0.35 (0.19)	0.37 (0.17)
Sponsored bills pre-primary	3.53 (3.75)	3.54 (4.42)	3.54 (4.32)	3.34 (4.02)
Cosponsored bills before primary	66.26 (42.63)	50.23 (39.22)	52.85 (40.17)	55.41 (44.90)
E-newsletters before primary	26.34 (40.30)	16.92 (16.38)	18.36 (21.90)	12.00 (11.08)
Roll-call participation before primary	0.90 (0.10)	0.91 (0.08)	0.91 (0.08)	0.94 (0.07)
Voting with party majority before primary	0.87 (0.10)	0.87 (0.10)	0.87 (0.10)	0.89 (0.09)
Voting with party leaders before primary	0.88 (0.10)	0.88 (0.11)	0.88 (0.11)	0.90 (0.09)

Note: Close primaries are those with margins smaller than 20 percentage points. Standard deviations are in the parentheses.

Columns (1) and (2) of Table 2 present a balance check between winners and losers in close primary elections within a 20-percentage-point vote margin. The two groups are well balanced on demographic and ideological characteristics and on the vote-based pre-primary behaviors (roll-call participation, voting with the party majority, voting with party leaders). The three count-based pre-primary activities (sponsored bills, cosponsored bills, e-newsletters) show a residual imbalance, with close losers somewhat more active than close winners in the pre-primary period. Because the dependent variable is a within-term post-primary proportion, a uniform level difference is already largely netted out; I additionally address the residual imbalance formally via the decomposed-RDD placebo (Table A.14) and the direct robustness check in subsection A.17.

3.2.2 External Validity of Regression Discontinuity Design

External validity warrants further consideration. The regression discontinuity design results are derived exclusively from legislators who faced competitive primary elections, which may limit their generalizability to the broader population of legislative representatives. To address this concern, I also estimate a difference-in-differences specification (Section 3.2.3) that uses the full sample and produces consistent results. As a more direct check on scope, I compare the contested-primary incumbents who enter the design against the non-challenged incumbents who fall outside it—the group that most concretely stands in for that broader population (subsection A.12). The two groups differ only modestly on observable characteristics: the two district-political variables of greatest concern, general-election margin and district presidential vote, differ by just 1.1 and 2.6 percentage points. More tellingly, in a three-group difference-in-differences comparison the non-challenged incumbents behave like narrow primary winners rather than losers across twelve effort and ideology outcomes—exactly what the lame-duck mechanism predicts for legislators who never carried a re-election threat to remove, and an indication that the post-primary effort decline is concentrated specifically among legislators whose re-election incentive disappears. The following considerations further suggest that findings from competitive primary elections may reasonably extrapolate to legislators in general, including those facing minimal or no primary challenges.

First, representatives in competitive and uncompetitive primary elections appear largely similar on descriptive characteristics. Columns (3) and (4) of Table 2 present legislators’ characteristics and behavior before the primary elections, distinguishing between those from close and non-close primary contests. Overall, the two groups appear similar across most observable variables, with e-newsletter counts showing the only statistically significant difference between the two groups.

Second, the motivations for primary challenges are similar across close and non-close contests. Using Cowburn (2024)’s classification of challenge motivations for U.S. House Representatives from 2006 to 2020 (following Boatright, 2014), I find that the underlying

reasons are largely similar across the two groups. Among close primary contests within 20% point margin, the most common reason for a challenge is the perception that the incumbent is too moderate (56%), followed by concerns over age and competence (17%), the incumbent being seen as too extreme (13%), scandals (5%), and national issues (4%). This distribution is not dramatically different from that of non-competitive primaries, where the most frequently cited reason remains the incumbent’s moderate ideology (45%), followed by age and competence (24%), the incumbent being perceived as too extreme (14%), national issues (4%), and scandals (0.7%).

Third, many close primary elections occur as a consequence of redistricting. Redistricting often disrupts an incumbent’s electoral advantages unexpectedly by forcing incumbents to run against one another or by splitting their loyal voter bases across districts. Because these disruptions often lie beyond the influence of legislators’ own strategic decisions or qualities, close primary contests in these instances are less likely to reflect systematic differences between legislators alone. The data confirm this pattern: on average, 40.14 House representatives face a primary election decided within a 20% point margin during a redistricting year, compared to 25.85 in a non-redistricting year. To the extent these challenges arise from unforeseen external shocks, the dynamics observed in close elections can reasonably generalize to legislators facing little or no primary competition; in a robustness check, I restrict the analyses to redistricting years only, with consistent results (Table A.3).

3.2.3 Difference-in-Differences Design

To demonstrate that the results are not a product of the small sample of close primary elections, I also employ a difference-in-differences design. In this research design, I compare individual legislators’ behavior across terms when they won primary elections versus terms when they lost. The advantage of this design is its ability to utilize the full sample, thereby representing a larger number of legislators and offering greater statistical power. As shown in the results, both designs yield similar conclusions, reinforcing the findings and providing

robust evidence.

The difference-in-differences specification differs across outcome types, reflecting the dependent-variable structure introduced in Section 3.1. For roll-call-based outcomes (roll-call participation, voting with the party majority, voting with party leaders, and roll-call extremism), I estimate legislators’ change in behavior after primary at the legislator-bill-vote level. Let Y_{ibt} denote legislator i ’s voting outcome on roll-call vote b in Congress t , PrimaryWin_{it} an indicator for whether the legislator won the primary in the Congress t covering vote b , and $\text{AfterPrimary}_{ibt}$ an indicator for whether the vote occurred after that primary’s election date. I estimate

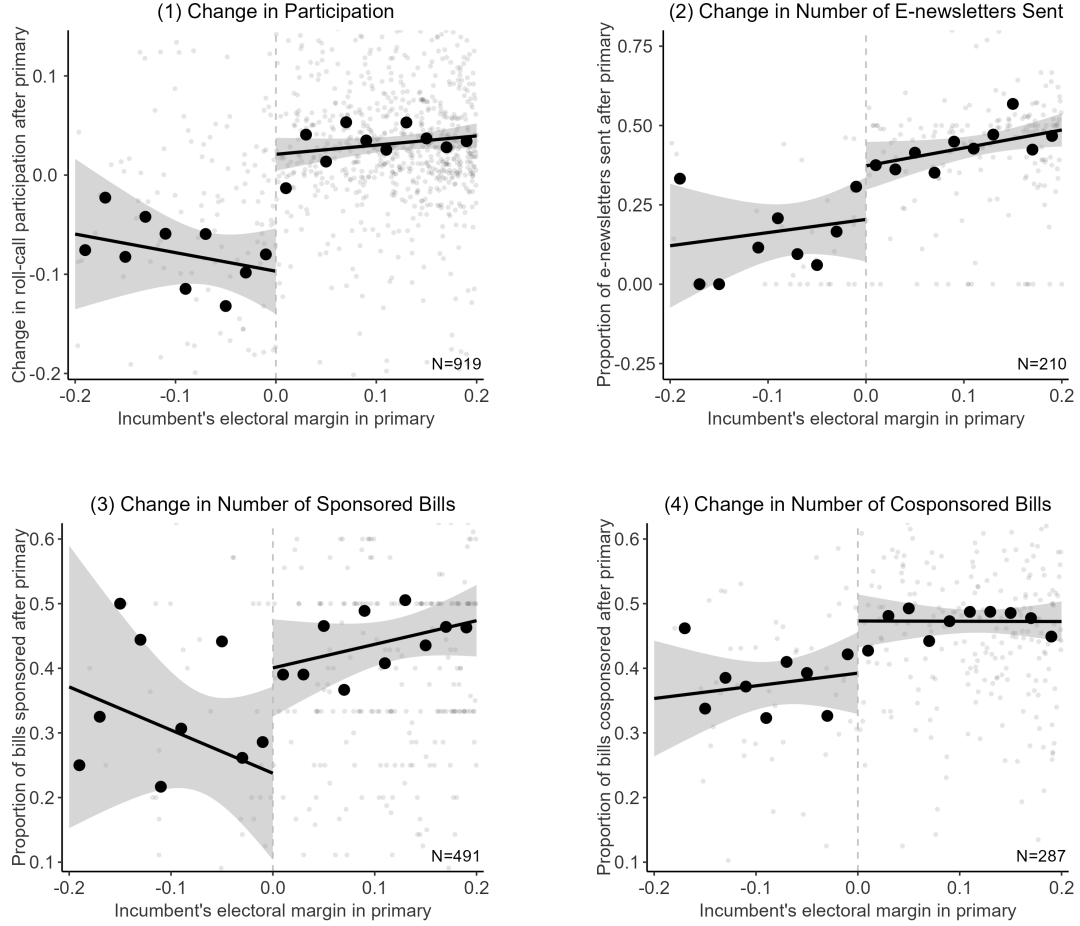
$$Y_{ibt} = \beta_0 + \beta_1 \text{PrimaryWin}_{it} + \beta_2 \text{AfterPrimary}_{ibt} + \beta_3 (\text{PrimaryWin}_{it} \times \text{AfterPrimary}_{ibt}) + \delta_i + \gamma_{bpt} + \varepsilon_{ibt}, \quad (4)$$

where δ_i denotes legislator fixed effects and γ_{bpt} denotes bill-by-party-by-Congress fixed effects (with p the legislator’s party), which absorb bill-level and party-specific shocks. Standard errors are clustered at the legislator level. The coefficient of interest is the interaction β_3 , which identifies the difference in the post-primary change in voting behavior between terms in which a legislator won versus lost the primary, net of time-invariant differences across legislators.

For non-vote outcomes (e-newsletters, sponsored bills, and cosponsored bills)—where legislators can more freely set their own agenda, as discussed in Section 3.1—the vote-level specification in equation (4) is not appropriate: these activities are each legislator’s own output rather than responses to a common set of bills, so there is no shared bill-level structure on which to define the γ_{bpt} fixed effects. For these outcomes I retain the aggregated legislator-period specification:

$$Y_{ipt} = \beta_0 + \beta_1 \text{PrimaryWin}_{it} + \delta_i + \gamma_{pt} + \varepsilon_{ipt}, \quad (5)$$

Figure 4: Regression Discontinuity Results on Legislators' Effort-related Activities



Note: Compared to legislators who marginally lose in primary elections, those who narrowly win exhibit an increase in various effort-related activities after the primaries. In the graphs, large black points represent averages computed within 0.02-point bins of the incumbent legislators' winning margins, while small gray points show the raw data. The lines display OLS fits estimated separately on each side of the threshold, and the y-axis indicates the change in legislators' behavior before and after the primary elections.

where Y_{ipt} is the legislator's within-term change in the outcome (the post-primary proportion in equation (1)), δ_i is legislator fixed effects, and γ_{pt} is congress-party fixed effects. The coefficient β_1 measures the effect of winning the primary on the legislator's change in behavior. All three non-vote outcomes remain positive and statistically significant whether or not legislator fixed effects are included. For the vote-based outcomes, equation (5) is also reported in the printed appendix (Table A.8) as a cross-check.

4 Primary Losers Exhibit Decreased Effort

Figure 4 visualizes the discontinuous difference in effort-related behaviors between primary losers and winners. In each panel, the x -axis is the running variable (incumbents' winning margin), the y -axis is the legislator's post-primary change in behavior, and the OLS fits are estimated separately on each side of the threshold at 0.

Panel (1) of Figure 4 illustrates the change in legislators' roll-call participation after primary elections. The plot shows a noticeable jump at the point of discontinuity; when legislators go from barely losing to barely winning a primary election, there is a significant increase in their roll-call participation rate in the post-primary periods. The other panels show similar discontinuous increases in e-newsletters sent and in bills sponsored and cosponsored.

Table 3 presents numerical estimates of how winning or losing a primary election influences legislators' effort-related activities under both research designs. The top panel reports the coefficients from the regression discontinuity design, and the bottom panel displays those from the difference-in-differences design. Both approaches show that winning a primary election is followed by a significant increase in effort-related behavior.

Legislators who lose their primary election become approximately 11 to 13 percentage points less likely to participate in roll-call votes. The other effort outcomes are measured as post-primary proportions, with 0.5 indicating no change; the table shows that losing a primary leads to a significant decline in e-newsletter activity, bill sponsorship, and cosponsorship, as reflected in all corresponding coefficients. As shown in the winner and loser mean rows, winners' means generally fall just below 0.5 and range from 0.474 to 0.439, suggesting their behavior remains relatively stable after the primary. In contrast, primary losers' means range from 0.377 to 0.171, well below 0.5. Across both designs, the estimates consistently indicate that legislators who lose a primary, and thus cannot seek reelection, substantially reduce effort in legislative work and constituent communication. A variety of placebo checks confirm this headline finding, including a placebo-window test showing that

Table 3: Numerical Results on Legislators' Effort-related Activities

Regression Discontinuity				
	Change in Roll-call Participation	Change in E-Newsletters	Change in Sponsorship	Change in Cosponsorship
Primary Win	0.129*** (0.019)	0.169** (0.072)	0.163** (0.077)	0.081* (0.042)
Observations	926	210	491	287
Winner Mean	0.033	0.439	0.445	0.473
Loser Mean	−0.091	0.171	0.284	0.377
Bandwidth	0.2	0.2	0.2	0.2
Specification	Linear	Linear	Linear	Linear
Difference in Differences				
	Change in Roll-call Participation	Change in E-Newsletters	Change in Sponsorship	Change in Cosponsorship
Primary Win	0.115*** (0.014)	0.204*** (0.070)	0.175** (0.073)	0.096*** (0.021)
Observations	3,622,179	863	2,189	1,321
Winner Mean	0.028	0.466	0.452	0.474
Loser Mean	−0.091	0.171	0.284	0.377
Specification	Vote-level	Aggregated	Aggregated	Aggregated
Fixed Effects	Legislator, Bill-Party-Cong.	Legislator, Party-Congress	Legislator, Party-Congress	Legislator, Party-Congress

* $p < 0.1$; ** $p < 0.05$; *** $p < 0.01$

Note: The difference-in-differences column for roll-call participation reports the $\text{PrimaryWin} \times \text{AfterPrimary}$ interaction from equation (4) estimated at the legislator–bill vote level with legislator and bill-by-party-by-Congress fixed effects; the remaining three columns report the PrimaryWin coefficient from equation (5) estimated at the legislator–period level. Winner Mean and Loser Mean are within-term changes after the primary: the change in the participation rate for roll-call participation (0 indicates no change), and the post-primary share post/(pre + post) for the other three columns (0.5 indicates no change). Difference-in-differences standard errors are clustered by legislator; regression-discontinuity standard errors are conventional OLS standard errors, with bootstrap standard errors reported as a robustness check in Table A.18. The numbers of observations are different across columns because the unit of observation differs and the data for each activity span different periods.

the effort effects are specific to the actual primary date rather than to an arbitrary short window, and a previous-Congress placebo showing that the same change is absent one term earlier (subsection A.16, subsection A.18, subsection A.19).

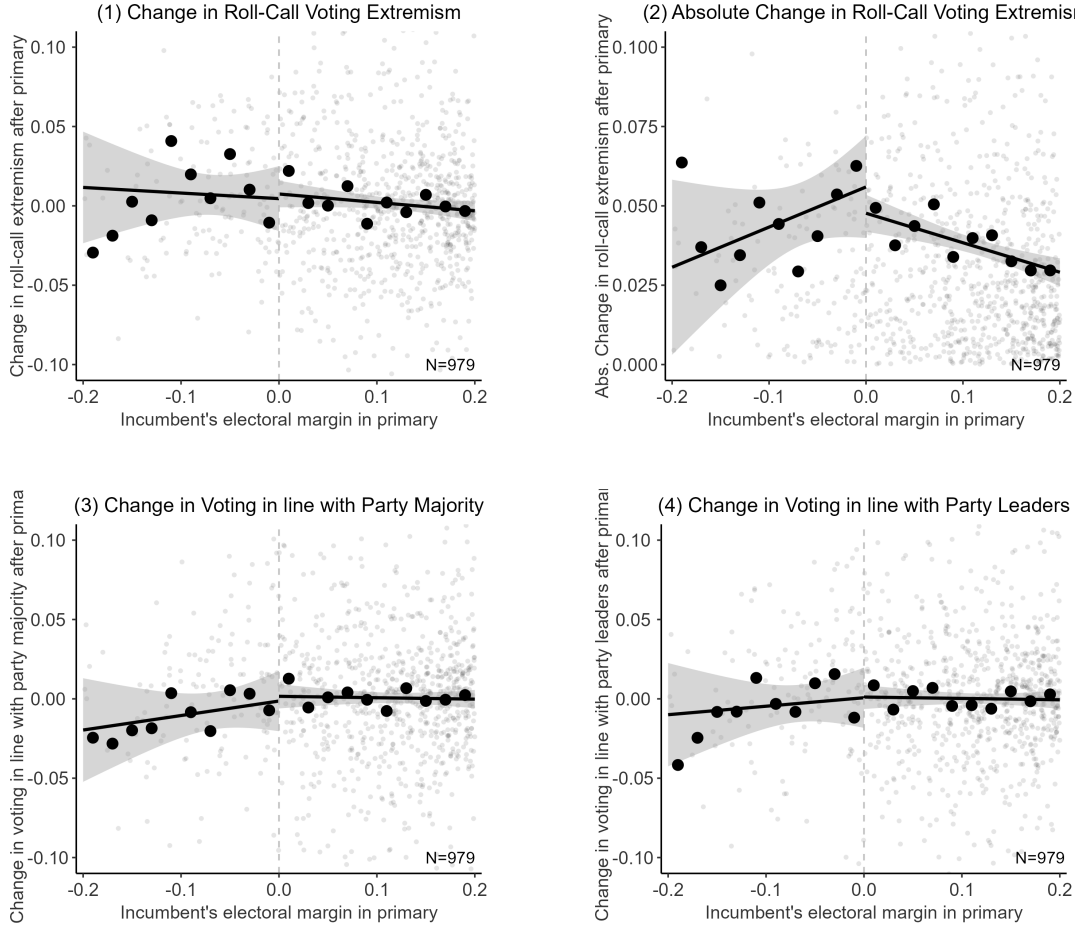
These findings closely align with conventional literature that argues legislators’ effort-related behavior is driven by their reelection motivation (Mayhew, 2004; Fenno, 1978; Cox and McCubbins, 2007). Conversely, it contrasts with predictions by scholars who contend that voters’ insufficient attention and political knowledge constrain representatives’ responsiveness and thus the effects of electoral incentives are limited (Achen and Bartels, 2017; Somin, 2016).

Nevertheless, lame duck legislators continue to participate in most roll-call votes and engage in various legislative activities, even after losing their primaries, albeit at a lower frequency than winners. After losing their primaries and until the general election, on average, these legislators still participate in 79% of roll-call votes, sponsor 2.67 bills, cosponsor 49.41 bills, and send 12.33 e-newsletters. These figures indicate that electoral incentives are hardly the sole driver of legislative activities. Their continued engagement may reflect ideological commitment, policy advocacy, reputation concerns for post-Congress careers, or a combination of these and other motivations.

5 Both Winners and Losers Show No Significant Shift in Policy Positions

Figure 5 visualizes the regression discontinuity results for legislators’ change in roll-call voting patterns: panels (1), (3), and (4) show changes in roll-call voting extremism, voting in line with the party majority, and voting in line with party leaders, respectively. In all three, there appears to be a negligible shift in the trend at the threshold, suggesting that primary election outcomes, and thus eligibility for the general election, do not significantly shift legislators’ post-primary policy positions.

Figure 5: Regression Discontinuity Results on Legislators' Policy Positions



Note: Legislators who narrowly win and lose do not show a statistically significant difference in their changes in roll-call voting after the primaries. In all measures, the changes for both winners and losers are statistically indistinguishable from zero, implying that neither group significantly alters their policy positions.

Table 4 presents the formal estimates, and the results from both designs consistently indicate negligible changes in policy positions. Across all four columns, the regression-discontinuity coefficients are at most 0.01 in magnitude, and the difference-in-differences estimates are likewise small, suggesting minimal differences in how winners and losers adjust their positions. The table also reports the mean changes in policy positions for winners and losers separately, and in every case, these mean changes are close to zero. This pattern indicates not only that winners and losers change their behavior similarly after primaries, but that neither group meaningfully alters its policy positions between pre- and post-primary

Table 4: Numerical Results on Change in Legislators’ Policy Positions

Regression Discontinuity				
	Change in Extremism	Absolute Change in Extremism	Change in Voting with Party Majority	Change in Voting with Party Leaders
Primary Win	0.003 (0.009)	−0.008 (0.006)	0.003 (0.008)	0.000 (0.008)
Observations	918	918	918	911
Winner Mean	0.001	0.037	0.001	0.000
Loser Mean	0.007	0.048	−0.007	−0.003
Bandwidth	0.2	0.2	0.2	0.2
Specification	Linear	Linear	Linear	Linear
Difference in Differences				
	Change in Extremism	Absolute Change in Extremism	Change in Voting with Party Majority	Change in Voting with Party Leaders
Primary Win	−0.006 (0.004)	−0.014** (0.006)	−0.002 (0.004)	−0.006 (0.004)
Observations	3,296,336	3,694	3,290,805	2,892,750
Winner Mean	−0.000	0.032	−0.000	0.001
Loser Mean	0.007	0.047	−0.007	−0.003
Unit of Analysis	Legislator–vote	Legislator–Congress	Legislator–vote	Legislator–vote
Fixed Effects	Legislator, Bill-Party- Cong.	Legislator, Party-Congress	Legislator, Bill-Party- Cong.	Legislator, Bill-Party- Cong.

* $p < 0.1$; ** $p < 0.05$; *** $p < 0.01$

Note: In the difference-in-differences panel, the “Primary Win” row reports the $\text{PrimaryWin} \times \text{AfterPrimary}$ interaction (equation (4); legislator and bill-by-party-by-Congress fixed effects) for the three vote-level columns and the PrimaryWin coefficient (equation (5); aggregated legislator-period, legislator and party-Congress fixed effects) for the absolute-change column, which has no vote-level analog. Winner Mean and Loser Mean give the average within-term change for terms won versus lost at the legislator–Congress level, computed on the ± 0.2 estimation sample in the regression-discontinuity panel and on the full sample in the difference-in-differences panel (0 indicates no change). For the extremism columns the dependent variable is the own-party-pole indicator extreme_{iv} . Difference-in-differences standard errors are clustered by legislator; regression-discontinuity standard errors are conventional OLS standard errors.

periods.

These results suggest limited effects of electoral concerns on legislators’ ideological po-

sitions, contrasting with the conventional view of position-taking as central to the electoral connection and competition (Mayhew, 2004; Downs, 1965). The null instead aligns with empirical evidence of rigidity in candidate positions (Hall, 2019; Rogers, 2017; Poole and Rosenthal, 2014; Lee, Moretti and Butler, 2004) and formal models that assume fixed policy positions (Alesina, 1988; Osborne and Slivinski, 1996).

One might worry that a null on the average shift masks offsetting individual-level movement: legislators within the same party could move in opposing ideological directions after primaries, leaving the aggregate unchanged even as individuals move substantially. The natural test is to examine absolute change in roll-call extremism, which captures movement in either direction without cancellation. Graph (2) of Figure 5 and the second column of Table 4 show a slight negative change on this measure, significant under the difference-in-differences design but not the regression discontinuity, offering weak evidence that primary losers change their voting patterns somewhat more than winners after primaries. These patterns are consistent with the last-period (shirking) literature, in which legislators no longer subject to a reelection constraint vote closer to their own ideological preferences (Kalt and Zupan, 1990; Zupan, 1990; Bender and Lott, 1996). The statistical significance is not robust across the two designs, however, so this interpretation remains speculative.

The null on policy positions itself, by contrast, is highly robust: the same result holds whether the dependent variable is the within-term regression coefficient β_{it} from equation (2) (RDD columns of Table 4), the vote-level own-party-pole, with-party, or with-leader indicators with roughly 2.9–3.3 million bill-vote observations each (DiD columns of Table 4), or a simple pre/post mean difference computed over equal-length windows without any regression adjustment (Table A.7). A final check for the RDD asks whether the null change could conceal a winner–loser gap that is already present before the primary and simply persists afterward, since a constant gap differences away to zero. I estimate separate RDDs on the pre-primary and the post-primary voting levels (subsection A.16): at the cutoff, the pre- and post-primary levels are small and at most marginally significant for both with-party

and with-leader voting. There is no meaningful gap before the primary and none after, so the null reflects genuine stability rather than a constant winner–loser gap that the change would difference away. The same roll-call data, by contrast, yields a statistically precise and substantively large effect on the effort margin of participation, indicating the null on policy positions reflects actual stability in legislators’ within-term voting rather than insensitivity of the measures.

6 Heterogeneous Effects of Electoral Incentives

6.1 Effort Effects Are More Substantial in Districts with Competitive General Elections

If primary winners exhibit more effort-related behavior than losers, it likely reflects strategic preparation for the upcoming general elections, where such activity can enhance their vote share. In some districts, however, the general election is expected to be minimally competitive, especially when a legislator or their party holds a dominant position. In such districts, legislators may worry less about reelection, and primary winners may face different incentives than those in more competitive environments (Burden, 2004). This suggests that the effort effect could depend on anticipated competition in the general election.

To test this, I divide the sample by whether the upcoming general election is expected to be competitive. Because incumbents in uncompetitive districts typically win by large margins, I use the incumbent’s vote share in the most recent general election: districts where the incumbent received more than 75% form the uncompetitive group, and those below form the competitive group.

While primary winners participate in roll-call voting more frequently than losers in both groups, Table 5 shows that the positive effort-related effects are more substantial and robust in competitive districts. In uncompetitive districts, several effort-related effects other than roll-call participation become insignificant or inconsistent across the two designs. These

Table 5: Results on Effort-related Behavior in Competitive and Uncompetitive Districts

Competitive Districts ($V_{t-1} < 0.75$)				
	Regression Discontinuity			
	Participation	E-Newsletters	Sponsored Bills	Cosponsored Bills
Primary Win	0.096*** (0.021)	0.218*** (0.082)	0.164* (0.087)	0.089* (0.049)
Observations	660	155	368	208
Bandwidth	0.2	0.2	0.2	0.2
Specification	Linear	Linear	Linear	Linear
	Difference in Differences			
	Participation	E-Newsletters	Sponsored Bills	Cosponsored Bills
Primary Win	0.116*** (0.015)	0.296*** (0.063)	0.252*** (0.087)	0.122*** (0.027)
Observations	2,643,524	635	1,600	948
Specification	Vote-level	Aggregated	Aggregated	Aggregated
Fixed Effects	Legislator, Bill-Party-Cong.	Legislator, Party-Congress	Legislator, Party-Congress	Legislator, Party-Congress
Uncompetitive Districts ($V_{t-1} > 0.75$)				
	Regression Discontinuity			
	Participation	E-Newsletters	Sponsored Bills	Cosponsored Bills
Primary Win	0.212*** (0.039)	-0.327* (0.190)	0.135 (0.178)	0.045 (0.081)
Observations	266	55	123	79
Bandwidth	0.2	0.2	0.2	0.2
Specification	Linear	Linear	Linear	Linear
	Difference in Differences			
	Participation	E-Newsletters	Sponsored Bills	Cosponsored Bills
Primary Win	0.107*** (0.033)	0.150*** (0.044)	-0.161 (0.229)	0.081 (0.054)
Observations	978,655	228	589	373
Specification	Vote-level	Aggregated	Aggregated	Aggregated
Fixed Effects	Legislator, Bill-Party-Cong.	Legislator, Party-Congress	Legislator, Party-Congress	Legislator, Party-Congress

* $p < 0.1$; ** $p < 0.05$; *** $p < 0.01$

Note: The top table reports competitive districts, where the incumbent's vote share in the previous general election was below 75%; the bottom table reports uncompetitive districts, where it was above 75%. The difference-in-differences and regression-discontinuity specifications are the same as in Table 3.

Difference-in-differences standard errors are clustered by legislator; regression-discontinuity standard errors are conventional OLS standard errors.

results suggest that electoral incentives, not non-electoral factors such as anticipated future service or committee assignments, are likely the primary driver.

Notably, the regression-discontinuity participation coefficient is higher in uncompetitive districts (0.212) than in competitive ones (0.096). This pattern is specific to the regression discontinuity, however: in the more highly powered difference-in-differences design the two are statistically indistinguishable (0.116 versus 0.107), and the uncompetitive regression-discontinuity estimate rests on a small close-primary sample of 266 legislators. I therefore read it cautiously. If it reflects a real difference, a plausible account is that a substantial share of legislators’ motivation to attend roll-calls is non-electoral, such as personal ideological commitments, consistent with the earlier finding that losers still take part in most roll-calls, albeit less frequently than winners; on this account, primary winners in competitive districts attend somewhat fewer roll-calls because they must devote more time to campaigning and visiting home districts ahead of the general election. But because the two designs disagree, the safer conclusion is that the participation effect is broadly comparable across district types, and it is the other effort outcomes that more clearly distinguish competitive districts.

6.2 Effects on Policy Positions Remain Null for Important Bills

The null on policy positions might stem from a narrower channel: legislators may respond to electoral incentives only on a small subset of salient or important bills. To test this, I restrict the analysis to bills deemed important.

I take three approaches to identifying “important” bills. First, I examine bills “whipped” by party leadership to test whether legislators respond to party pressure. Party leaders occasionally conduct whip counts — informal surveys on significant agenda items — typically reserved for high-priority bills. If electoral incentives operate through party pressure, the effects should be more pronounced on whipped bills. This analysis leverages whip count data from the Democratic Party (1955–1986) and the Republican Party (1975–1980) provided by Evans (2012).

Second, I focus on bills previous scholars deemed especially significant or salient. I pool two standard measures of legislative importance, each linked to specific bill numbers: priority agenda items from Curry and Lee (2020) and landmark enactments compiled from CQ Press.² I apply the same measure of position change to these bills. If legislators are more responsive to high-profile legislation, the effects should be more pronounced on them than on the full sample.

Third, I focus on contested bills — those close to passing or failing. Such bills are especially informative because each legislator is more likely to cast a pivotal vote. If electoral incentives heighten legislators’ attention to legislative outcomes, the effects should be more pronounced on these narrowly decided bills. To operationalize this, I calculate the difference between yea and nay votes for each bill and repeat the analysis for bills in the lowest 30% of that difference (fewer than 62 votes between yea and nay).³

²The priority-agenda series covers only Congresses 103–115 and supplies too few within-window votes per legislator to support the regression-discontinuity change-score once the minimum-vote requirement below is imposed; the landmark series spans Congresses 84–113 and supplies the needed coverage. The two measures are largely redundant, so the results are nearly identical using the landmark series alone, while the priority-agenda series alone is underpowered.

³30% is arbitrary; results are unchanged at the 20% or 50% thresholds. This measure of “contestedness” is posterior, capturing realized vote outcomes after electoral incentives may have influenced legislators’ votes, but realized contestedness likely approximates expected contestedness.

Table 6: Effects of Winning Primary on Voting on Important Bills

	Regression Discontinuity			Difference-in-Differences		
	Whipped Bills	Significant/ Landmark Bills	Contested Bills	Whipped Bills	Significant/ Landmark Bills	Contested Bills
Change in Roll-call Extremism	−0.051 (0.075)	−0.013 (0.022)	0.000 (0.016)	−0.072 (0.048)	−0.015** (0.008)	−0.010 (0.006)
Change in Voting with Party Majority	−0.046 (0.081)	−0.012 (0.021)	0.010 (0.016)	−0.029 (0.047)	−0.014* (0.008)	−0.010 (0.006)
Change in Voting with Party Leaders	−0.137 (0.086)	0.000 (0.023)	−0.004 (0.018)	−0.078 (0.049)	−0.018* (0.009)	−0.015** (0.007)
Change in Roll-call Participation	0.166*** (0.051)	0.119*** (0.025)	0.128*** (0.019)	0.162*** (0.036)	0.094*** (0.018)	0.109*** (0.013)
Observations	124–148	656–732	899–919	28k–36k	496k–633k	942k– 1,127k
Specification	Bandwidth = 0.2 Specification = Linear			Legislator + Bill-Party-Congress FE		

* $p < 0.1$; ** $p < 0.05$; *** $p < 0.01$

Note: Each cell reports the effect of winning the primary on the change in the respective voting behavior; standard errors are in parentheses and clustered by legislator in the difference-in-differences design. The difference-in-differences and regression-discontinuity specifications are the same as in Table 3, and the roll-call extremism outcome is the own-party-pole indicator extreme_{iv} defined in subsection 3.1. The three bill subsamples are whipped bills (Evans 2012; Congresses 84–99), significant/landmark bills (priority agenda items from Curry & Lee 2020 pooled with landmark enactments compiled from CQ Press; Congresses 84–115), and contested bills (bottom 30% of $|\text{yea} - \text{nay}|$; Congresses 84–117). To estimate a legislator’s within-subsample position or participation change, the regression-discontinuity columns require at least five subsample votes on each side of the primary (cast votes for the position outcomes, available roll calls for participation; Appendix A.3); the difference-in-differences columns pool all votes and impose no such requirement. Regression-discontinuity sample sizes therefore differ across rows (positions versus participation) and are reported as ranges; vote-level sizes likewise differ within each column due to outcome-specific missingness. Because the difference-in-differences samples for the significant/landmark and contested columns exceed half a million votes, coefficients there can be statistically distinguishable from zero while remaining substantively negligible in magnitude ($|\hat{\beta}| < 0.02$).

Each cell in Table 6 reports the effect of winning a primary on subsequent voting for whipped, significant/landmark, and contested bills. Across both designs, coefficients for roll-call extremism and voting alignment (with party majority or party leaders) are small and substantively indistinguishable from zero. A few of the difference-in-differences position

estimates reach conventional significance in the significant/landmark and contested columns, but only because those pooled vote-level samples exceed half a million observations; the coefficients themselves are negligible, all below 0.02 in absolute value. This pattern reinforces the earlier evidence on the rigidity of legislators’ policy positions: primary winners and losers do not systematically differ in how they adjust positions on major votes after primary elections. The bottom row provides a power check. On all three subsamples—each restricted to legislators with at least five available roll calls per side, so the change-scores are well estimated—roll-call participation rises sharply and significantly for primary winners, so the flat policy positions reflect genuine stability rather than a lack of statistical power. Nor does the null stem from losers disengaging—their participation falls only about nine percentage points after the primary while winners’ is essentially unchanged (Table 3, Winner/Loser Means), leaving them active on a substantial share of votes.

7 Conclusion

Electoral accountability lies at the heart of democratic governance, compelling representatives to remain responsive to constituents. Alexander Hamilton underscored this during the debates over term limits, arguing that removing the prospect of reelection would diminish legislators’ motivation: “One ill effect of the exclusion [of reelection] would be a diminution of the inducements to good behavior” (Hamilton, 2003).

Despite the recognized importance of electoral incentives, identifying precisely which “good behavior” they induce remains empirically challenging. Representatives strategically choose whether to seek reelection or retire, making it difficult to isolate the influence of electoral incentives from selection bias. This study addresses this challenge by leveraging the post-primary period, when incumbents face a critical juncture: continuing their reelection campaigns or exiting after a primary defeat. Primary winners, still in electoral contention, exert greater legislative effort than primary losers, including more active roll-call participa-

tion, bill sponsorship, and constituent communication. Yet legislators' policy positions in roll-call voting do not significantly change, whether they won the primary or not.

Taken together, the results show that electoral incentives discipline legislative effort but not legislative position-taking. The effort findings reinforce classic theories of electoral accountability (Barro, 1973; Ferejohn, 1986; Mayhew, 2004): legislators invest more in participation, sponsorship, and constituent communication when a general election still looms, and they do so most where that election is expected to be competitive. Contrary to the worry that voters' inattention blunts accountability (Achen and Bartels, 2017), representatives respond to the prospect of competition itself.

Their roll-call positions, by contrast, barely change. Although position-taking is classically treated as central to the electoral connection (Mayhew, 2004) and electoral competition (Downs, 1965), positions hold steady after a primary defeat, even when electoral pressure should produce the sharpest response: whipped bills, significant agenda items, and narrowly contested ones. I interpret this divergence as reflecting a difference in kind between the two margins. Effort is an individually controlled, observable signal that can pay off in the next campaign; roll-call positions are anchored by more durable commitments—members' own ideological convictions and the established policy records they were elected on (Poole and Rosenthal, 2014; Hall, 2019; Lee et al., 2004; Alesina, 1988; Osborne and Slivinski, 1996)—that the lifting of reelection pressure does not dislodge.

The same stability informs how scholars read roll-call data. Parties control resources legislators need to win (Cox and McCubbins, 2007; Krehbiel, 1993), yet party-line voting is unchanged when a member's reelection incentive vanishes, so within-term electoral pressure is not the primary channel through which parties pull members into line. For the same reason, because ideological scores such as DW-NOMINATE (Poole and Rosenthal, 1985; McCarty, 2016) are stable across the primary cutoff even on the strongest-prediction bills, they are unlikely to be mere artifacts of immediate electoral pressure operating through party discipline or donors (Snyder and Groseclose, 2000; Bronars and Lott, 1997).

The findings also bear on the roots of polarization. Because sitting incumbents do not shift their positions in their term’s closing months, regardless of primary outcome, widening partisan divides are more plausibly driven by who enters and leaves office—replacement, sorting, and nomination—than by repositioning among those already in office (Moskowitz et al., 2024).

This null is subject to one qualification: it describes a specific margin, not elections in general. The design isolates the within-term effect of removing future reelection incentives; it does not preclude electoral incentives from shaping policy at a slower pace over a longer time span. Albouy (2011) finds that U.S. Senators, who serve six-year terms, moderate their voting more in the years closer to reelection, and Caughey and Warshaw (2018) document multi-year responsiveness of state policy to public opinion—evidence that electoral pressure may accumulate over a longer horizon. A null effect over a relatively short period in this design does not necessarily contradict accounts in which electoral pressure builds slowly enough that its abrupt removal over a few months leaves recorded votes unchanged. Nor can the design detect positioning that adjusted *before* the primary—a member casting tough votes in anticipation of not running again—or change deterred by the very threat of a challenge; the null accordingly bounds the post-primary, within-term channel rather than every path by which electoral competition could discipline positions.

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A Appendix

A.1 Post-congress Career Trajectory of Primary Losers

Table 1 in the main article presents the post-Congress career trajectories of primary election losers. It is notable that most legislators do not pursue elected office again after losing a primary election although these are ex-post realized decisions and do not directly reveal candidates' true incentives during their lame-duck periods. To probe this as far as the data allow, I conduct a separate analysis excluding all primary losers who later sought an elected office. The results remain consistent, as shown in Table A.1.

Table A.1: Results without Re-running Primary Losers

Effort Behavior				
	Regression Discontinuity			
	Participation	E-Newsletters	Sponsored Bills	Cosponsored Bills
Primary Win	0.132*** (0.020)	0.169** (0.077)	0.162* (0.089)	0.091* (0.050)
Observations	897	203	473	275
Bandwidth	0.2	0.2	0.2	0.2
Specification	Linear	Linear	Linear	Linear
	Difference in Differences			
	Participation	E-Newsletters	Sponsored Bills	Cosponsored Bills
Primary Win	0.106*** (0.021)	0.216*** (0.083)	0.189** (0.087)	0.089*** (0.026)
Observations	3,685	856	2,171	1,309
Specification	Aggregated	Aggregated	Aggregated	Aggregated
Fixed Effects	Legislator, Party-Cong.	Legislator, Party-Cong.	Legislator, Party-Cong.	Legislator, Party-Cong.
Policy Position				
	Regression Discontinuity			
	Extremism	Abs. Extremism	With Party	With Leader
Primary Win	0.003 (0.010)	-0.010 (0.007)	0.004 (0.009)	0.002 (0.009)
Observations	889	889	889	882
Bandwidth	0.2	0.2	0.2	0.2
Specification	Linear	Linear	Linear	Linear
	Difference in Differences			
	Extremism	Abs. Extremism	With Party	With Leader
Primary Win	0.010 (0.009)	-0.014* (0.007)	0.009 (0.009)	0.003 (0.009)
Observations	3,665	3,665	3,665	3,605
Specification	Aggregated	Aggregated	Aggregated	Aggregated
Fixed Effects	Legislator, Party-Cong.	Legislator, Party-Cong.	Legislator, Party-Cong.	Legislator, Party-Cong.

* $p < 0.1$; ** $p < 0.05$; *** $p < 0.01$

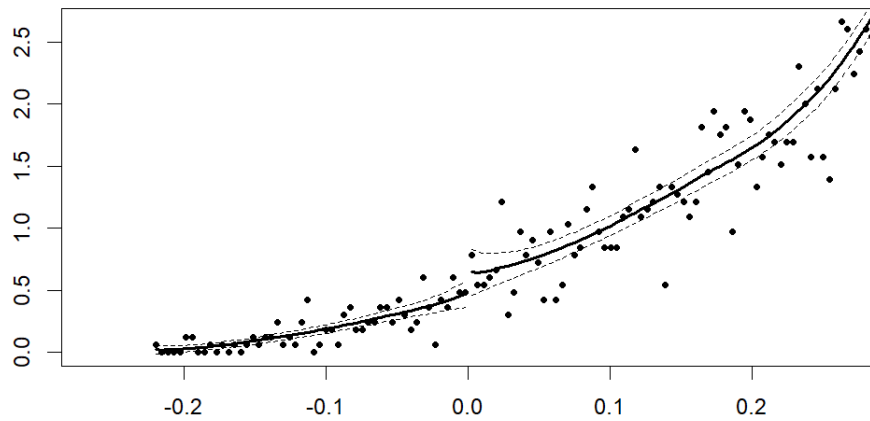
Note: These results are based on the sample without any House representatives who lost in primary elections but ran for an election later. The results are mostly the same as the main findings.

A.2 Manipulation Test

The McCrary density test is used in regression discontinuity designs to check for potential manipulation of the running variable around the threshold (McCrary, 2008). The idea is that if individuals can manipulate the running variable, there might be a discontinuity in its density function at the cutoff, indicating possible sorting or manipulation. The test produces

Figure A.1; the estimated log difference in the height of the running-variable density at the cutoff is 0.30 (SE = 0.23), statistically indistinguishable from zero ($p = 0.18$). This suggests that manipulation of the running variable is unlikely.

Figure A.1: Plot from McCrary Test



A.3 Excluded Legislators

The roll-call sample begins with 14,554 Congress-legislators whose primary election can be matched to a roll-call voting record. I drop 241 who participated in fewer than half of their seat’s roll-call votes in their respective Congresses, and, because the research designs require voting behavior both before and after the primary, a further 635 with fewer than ten recorded votes on one side. Almost all of these fall short on the post-primary side (443 cast no post-primary votes at all, the rest only a handful), for two distinct reasons. About two-thirds are concentrated in the 1956, 1958, and 1960 cycles, when Congress adjourned by late summer, before those states’ largely September primaries, so that no post-primary roll calls existed to observe; once sessions began running into the fall (from 1962 onward) this case nearly disappears. The remainder are members who left office before the primary (through resignation, death, or appointment to other office) or who, though still serving, cast nothing in the brief window between an August primary and the autumn campaign recess. Only a small number fall short on the pre-primary side instead (11 with none at all), almost all special-election winners who entered after the primary. Because those dropped for sparse post-primary voting are, by construction, among the least active after the primary, their exclusion is more likely to understate than to inflate the post-primary participation decline the analysis identifies. These restrictions leave 13,678 Congress-legislators, the panel underlying both the aggregated legislator-period and vote-level specifications.

The same principle applies, at the subsample level, to the analyses of important bills (Section 6.2). A legislator’s change in position or participation *on a particular set of bills* can only be estimated when enough such votes fall on each side of the primary; when only one or two do, the change-score is mechanically driven toward ± 1 and a handful of these cells can dominate the estimate. I therefore require at least five subsample votes per side for the regression-discontinuity change-scores in those analyses—five rather than the full panel’s ten because the bill subsamples are far sparser (whipped and landmark votes are a small fraction of all roll calls), so the lower threshold removes the unestimable one- and two-vote cells while retaining adequate power. Positions are filtered on cast votes and participation on available roll calls, mirroring the full-panel rule. The pooled vote-level difference-in-

differences estimates do not form per-legislator cells and are left unfiltered.

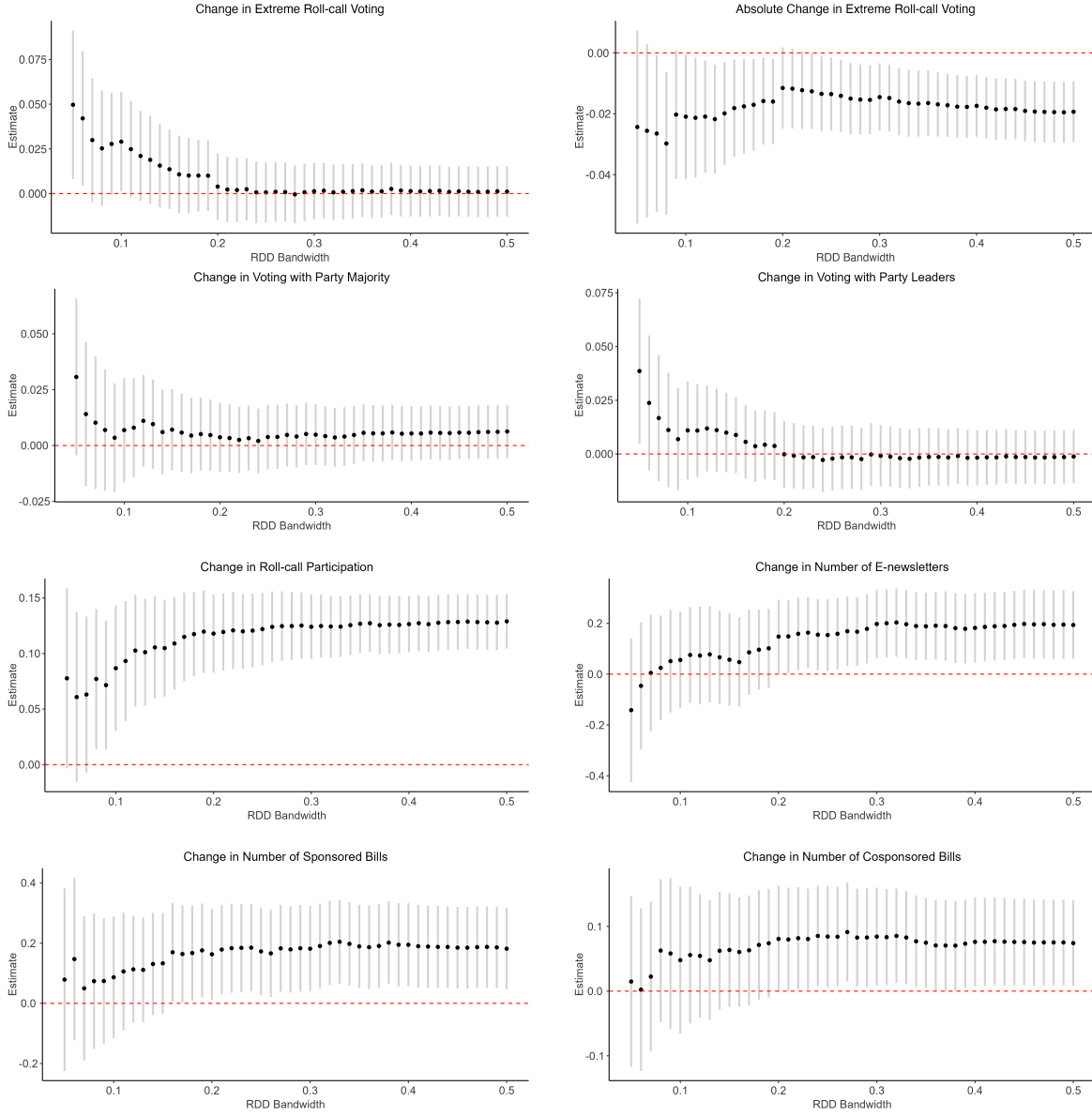
A.4 Regression Discontinuity Results with Different Bandwidths

The choice of bandwidth in a regression discontinuity design can be arbitrary, so it is important to demonstrate that conclusions are not driven by an particular choice of the specification. Figure A.2 shows the RDD estimate over different sizes of the bandwidths. The results are rather unstable for very narrow bandwidths due to the small sample sizes.

Due to the shorter time spans of the data for e-newsletters, sponsorship, and cosponsorship, the sample sizes are limited when using narrow bandwidths. As a result, many coefficients are statistically insignificant and fluctuate with small adjustments in bandwidth sizes within these narrow ranges. Nonetheless, the signs of the coefficients generally remain largely stable and align with the main conclusion of this article. Where the sample sizes are sufficiently large, the coefficients are largely consistent with the results from the difference-in-differences design.

Data-driven bandwidth selection has become a standard part of regression discontinuity practice. Table A.2 therefore reports estimates obtained with the mean-squared-error-optimal bandwidth of Calonico, Cattaneo and Titiunik (2014), using a local-linear specification with a uniform kernel. The optimal bandwidths are narrow—roughly $h = 0.04$ to 0.08 —which leaves very few observations near the cutoff: the effective samples for the effort outcomes range from about 190 for participation down to fewer than 30 for e-newsletters, whose data cover the shortest span. At these bandwidths most coefficients are no longer statistically significant, and the participation estimate (0.05) is smaller than its ± 0.2 counterpart (0.129); the e-newsletter estimate rests on only 26 observations and is too imprecise to interpret, even reversing sign. The policy-position estimates remain statistically insignificant at conventional levels throughout, with the sole exception of the with-leader cell, which is marginal at the 10% level on only 157 observations. These results reinforce the bandwidth-sensitivity pattern in Figure A.2: the close-primary discontinuity is identified from a genuinely small number of competitive races, and the design’s statistical power rests on the

Figure A.2: RDD Estimates from Different Bandwidths



Note: Each graph visualizes the RDD estimates of changes in legislator behavior over different bandwidth. The x -axis in each graph is for the bandwidth, and the y -axis for the estimate size. The vertical gray lines for each bandwidth represents 95% confidence interval.

wider window used in the main analysis. I therefore report the optimal-bandwidth estimates transparently as a limitation of the close-primary RD rather than as corroborating evidence.

Table A.2: RD Estimates at the MSE-Optimal Bandwidth

	Coefficient	Robust SE	<i>p</i> -value	Optimal <i>h</i>	<i>N</i>
<i>Effort-related behavior</i>					
Change in Roll-call Participation	0.051	0.048	0.474	0.048	186
Change in E-Newsletter Activity	−0.180	0.366	0.452	0.035	26
Change in Bill Sponsorship	0.196	0.188	0.236	0.043	79
Change in Bill Cosponsorship	−0.010	0.078	0.778	0.058	66
<i>Policy positions</i>					
Change in Roll-call Extremism	0.031	0.027	0.153	0.068	250
Absolute Change in Roll-call Extremism	−0.022	0.025	0.319	0.054	204
Change in Voting with Party Majority	0.008	0.026	0.778	0.081	302
Change in Voting with Party Leader	0.052*	0.034	0.059	0.041	157

Note: Estimates use the mean-squared-error-optimal bandwidth of Calonico et al. (2014), implemented with `rdrobust` (local-linear, uniform kernel). “Coefficient” is the conventional RD point estimate; “Robust SE” and “*p*-value” are the bias-corrected robust standard error and *p*-value. *N* is the number of observations within the optimal bandwidth. **p* < 0.10, ***p* < 0.05, ****p* < 0.01.

A.5 Regression Discontinuity Results from Redistricting Years

The regression discontinuity design only derives local average treatment effect and is not externally valid. The specific concern is that those legislators who are subject to close primary elections might be systematically and significantly different from the other legislators who face non-close primaries or no primary challenges at all. To address this concern indirectly, I implement the same research design focusing on only redistricting years, assuming that in redistricting years, legislators are more likely to face substantial primary competition in an as-if random fashion. It suffers from fewer sample and weaker statistical power as it includes fewer election years, but the results in Table A.3 show that the sizes and direction of the coefficients remain mostly consistent with the original findings.

Table A.3: Regression Discontinuity Results from Redistricting Years

	Change in Roll-call Participation	Change in E-Newsletter	Change in Sponsorship	Change in with Cosponsorship
Primary Win	0.123*** (0.029)	0.131 (0.113)	0.052 (0.146)	0.074 (0.056)
Observations	280	73	124	111
Bandwidth	0.2	0.2	0.2	0.2
Specification	Linear	Linear	Linear	Linear

	Change in Roll-call Extremism	Absolute Change in Roll-call Extremism	Change in Voting with Party Majority	Change in Voting with Party Leaders
Primary Win	0.024 (0.015)	-0.008 (0.011)	0.011 (0.012)	0.002 (0.013)
Observations	279	279	279	277
Bandwidth	0.2	0.2	0.2	0.2
Specification	Linear	Linear	Linear	Linear

* $p < 0.1$; ** $p < 0.05$; *** $p < 0.01$

Note: Standard errors are in the parentheses and are conventional OLS standard errors. Each column is a separate local-linear regression discontinuity estimate at bandwidth ± 0.2 restricted to redistricting-cycle years (primary years 1962, 1972, 1982, 1992, 2002, 2012, 2022).

A.6 Policy Position Change in Competitive and Uncompetitive Districts

In the main article, I assess whether the effects of effort-related behavior vary with the expected competitiveness of the general election, finding that these effects are more pronounced in districts where primary winners anticipate competitive general elections. To examine whether similar heterogeneous effects exist for legislators' policy positions, I compare changes in policy positions after the primary between competitive and uncompetitive districts.

Table A.4 shows estimates of legislators' changes in policy positions in both competitive and uncompetitive districts following the primary elections. Across both research designs, the directional measures—signed extremism, voting with the party majority, and voting

with party leaders—show no significant changes in either group. The absolute change in roll-call extremism is the one exception: winners’ positions are more stable than losers’ in uncompetitive districts (-0.038 , $p < 0.01$ in the regression discontinuity; -0.028 , $p < 0.10$ in the difference-in-differences), the same winner-stability pattern documented for the full sample (cf. the -0.014 difference-in-differences estimate in Table 4), here concentrated where general-election pressure is weakest. Directional position-taking, by contrast, remains unaffected by the expected competitiveness of the upcoming general election, reinforcing the argument that adjusting policy stances is unlikely to be a predominant strategy for legislators seeking reelection.

Table A.4: Changes in Policy Positions in Competitive and Uncompetitive Districts

Competitive Districts ($V_{t-1} < 0.75$, $N = 2,706$)				
	Regression Discontinuity			
	Change in Roll-call Extremism	Absolute Change in Roll-call Extremism	Change in Voting With Party Majority	Change in Voting With Party Leaders
Primary Win	0.005 (0.010)	-0.000 (0.007)	0.006 (0.009)	0.000 (0.009)
Observations	653	653	653	649
Bandwidth	0.2	0.2	0.2	0.2
Specification	Linear	Linear	Linear	Linear
	Difference in Differences			
	Change in Roll-call Extremism	Absolute Change in Roll-call Extremism	Change in Voting With Party Majority	Change in Voting With Party Leaders
Primary Win	0.004 (0.009)	0.000 (0.006)	0.011 (0.007)	-0.001 (0.007)
Observations	2,705	2,705	2,705	2,662
Specification	Aggregated	Aggregated	Aggregated	Aggregated
Fixed Effects	Legislator, Party-Cong.	Legislator, Party-Cong.	Legislator, Party-Cong.	Legislator, Party-Cong.
Uncompetitive Districts ($V_{t-1} > 0.75$, $N = 989$)				
	Regression Discontinuity			
	Change in Roll-call Extremism	Absolute Change in Roll-call Extremism	Change in Voting With Party Majority	Change in Voting With Party Leaders
Primary Win	0.000 (0.019)	-0.038*** (0.014)	-0.001 (0.017)	0.008 (0.017)
Observations	265	265	265	262
Bandwidth	0.2	0.2	0.2	0.2
Specification	Linear	Linear	Linear	Linear
	Difference in Differences			
	Change in Roll-call Extremism	Absolute Change in Roll-call Extremism	Change in Voting With Party Majority	Change in Voting With Party Leaders
Primary Win	0.016 (0.018)	-0.028* (0.014)	-0.009 (0.023)	-0.002 (0.018)
Observations	989	989	989	972
Specification	Aggregated	Aggregated	Aggregated	Aggregated
Fixed Effects	Legislator, Party-Cong.	Legislator, Party-Cong.	Legislator, Party-Cong.	Legislator, Party-Cong.

* $p < 0.1$; ** $p < 0.05$; *** $p < 0.01$

Note: The top table shows the results from competitive districts where the incumbent legislator's previous vote share was less than 75%. The bottom table shows the results from less competitive districts where the previous vote share was greater than 75%.

A.7 Results Over Different Periods

An additional analysis was conducted to examine whether the effects vary across different time periods. Given the limited sample size, it was not feasible to segment the data into very fine time intervals, and I divided the sample into two periods: before and after 1994, which marks the beginning of a significant rise in ideological polarization and stricter party discipline.

The results in Table A.5 show that the main policy-position conclusions are stable across both periods: changes in roll-call extremism and party-line voting are small and statistically insignificant in nearly all cells, with only a few marginal exceptions that do not overturn the overall null on policy positions.

Table A.5: Policy Position Results in the Periods Before and After 1994

After 1994 ($N = 1,566$)				
	Regression Discontinuity			
	Change in Roll-call Extremism	Absolute Change in Roll-call Extremism	Change in Voting With Party Majority	Change in Voting With Party Leaders
Primary Win	-0.009 (0.008)	0.000 (0.006)	0.000 (0.009)	0.001 (0.008)
Observations	360	360	360	358
Bandwidth	0.2	0.2	0.2	0.2
Specification	Linear	Linear	Linear	Linear
	Difference in Differences			
	Change in Roll-call Extremism	Absolute Change in Roll-call Extremism	Change in Voting With Party Majority	Change in Voting With Party Leaders
Primary Win	-0.004 (0.007)	-0.006* (0.003)	0.014** (0.006)	0.010* (0.006)
Observations	1,566	1,566	1,566	1,537
Specification	Aggregated	Aggregated	Aggregated	Aggregated
Fixed Effects	Legislator, Party-Cong.	Legislator, Party-Cong.	Legislator, Party-Cong.	Legislator, Party-Cong.
Before 1994 ($N = 2,128$)				
	Regression Discontinuity			
	Change in Roll-call Extremism	Absolute Change in Roll-call Extremism	Change in Voting With Party Majority	Change in Voting With Party Leaders
Primary Win	0.010 (0.013)	-0.013 (0.009)	0.005 (0.012)	0.001 (0.012)
Observations	558	558	558	553
Bandwidth	0.2	0.2	0.2	0.2
Specification	Linear	Linear	Linear	Linear
	Difference in Differences			
	Change in Roll-call Extremism	Absolute Change in Roll-call Extremism	Change in Voting With Party Majority	Change in Voting With Party Leaders
Primary Win	0.013 (0.011)	-0.020** (0.010)	0.007 (0.013)	-0.004 (0.012)
Observations	2,128	2,128	2,128	2,097
Specification	Aggregated	Aggregated	Aggregated	Aggregated
Fixed Effects	Legislator, Party-Cong.	Legislator, Party-Cong.	Legislator, Party-Cong.	Legislator, Party-Cong.

* $p < 0.1$; ** $p < 0.05$; *** $p < 0.01$

Note: Comparison of changes in policy positions before and after 1994. The top panel covers election years after 1994 (from 1996 onward); the bottom panel covers election years through 1994.

A.8 Legislators’ Anticipation of Presidential Appointment

A potential concern for primary losers is non-electoral career prospects, such as presidential appointments to federal or local office (Jenkins and Nokken, 2008). If losers anticipate such appointments in exchange for adhering to the party line, their policy positions could remain stable for reasons unrelated to electoral incentives—making the null on policy positions an artifact of substitute motivations, not evidence against the electoral-incentives channel.

To test this, I restrict the analysis to legislators in the president’s party during midterm-election years. Because the president serves at least two more years and rarely appoints across party lines, these lawmakers are substantially more likely than others to anticipate a post-Congress appointment. If the appointment-anticipation channel drives the null on policy positions, these legislators should show the smallest post-primary changes. The results (Table A.6) are mixed: the regression discontinuity design shows three statistically significant shifts (winners are 3.4 percentage points more likely to cast an extreme roll-call vote, show a 3.7-point smaller absolute extremity shift, and are 3.5 points more likely to vote with party leaders), but the difference-in-differences estimates are small and statistically insignificant. The pattern does not strongly support the appointment-anticipation channel as the driver of the null on policy positions.

Table A.6: Policy Positions Among Legislators in the President’s Party in Midterm Years

	Regression Discontinuity			
	Change in Roll-call Extremism	Absolute Change in Roll-call Extremism	Change in Voting With Party Majority	Change in Voting With Party Leaders
Primary Win	0.034* (0.019)	−0.037*** (0.014)	0.031 (0.019)	0.035** (0.018)
Observations	226	226	226	226
Bandwidth	0.2	0.2	0.2	0.2
Specification	Linear	Linear	Linear	Linear
	Difference in Differences			
	Change in Roll-call Extremism	Absolute Change in Roll-call Extremism	Change in Voting With Party Majority	Change in Voting With Party Leaders
Primary Win	−0.015 (0.043)	−0.025 (0.029)	−0.007 (0.037)	0.009 (0.033)
Observations	889	889	889	878
Specification	Aggregated Legislator	Aggregated Legislator, Party-Cong.	Aggregated Legislator, Party-Cong.	Aggregated Legislator, Party-Cong.
Fixed Effects				

* $p < 0.1$; ** $p < 0.05$; *** $p < 0.01$

Note: These results are from legislators who face primary elections during midterm election years and belong to the president’s party. These legislators are more likely than others to anticipate a post-Congress appointment by the president, as he will remain in office for at least two more years and rarely appoints legislators from the opposing party. The difference-in-differences estimate for Change in Roll-call Extremism includes legislator fixed effects only (no Party-Congress fixed effects).

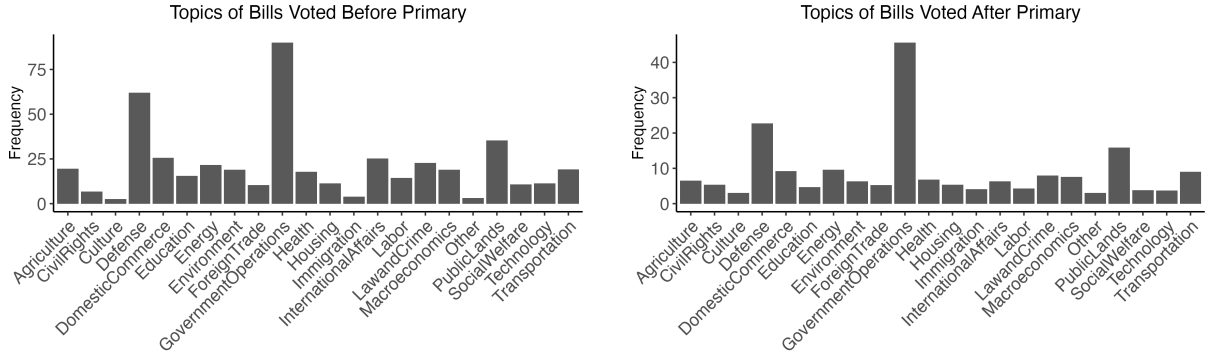
A.9 Descriptive Bill Differences Before and After Primary Elections

In the main article, I account for potential differences in the types of bills voted before and after primary elections by using regression models with bill fixed effects, thereby isolating changes in legislators’ voting behavior from potential variations in bill content.

Nevertheless, to ensure that the composition of bills is not dramatically different before and after primary elections, I conduct several descriptive comparisons, examining the distribution of bill topics and legislators’ general voting patterns.

Figure A.3 presents the average number of topics of bills voted on per congressional

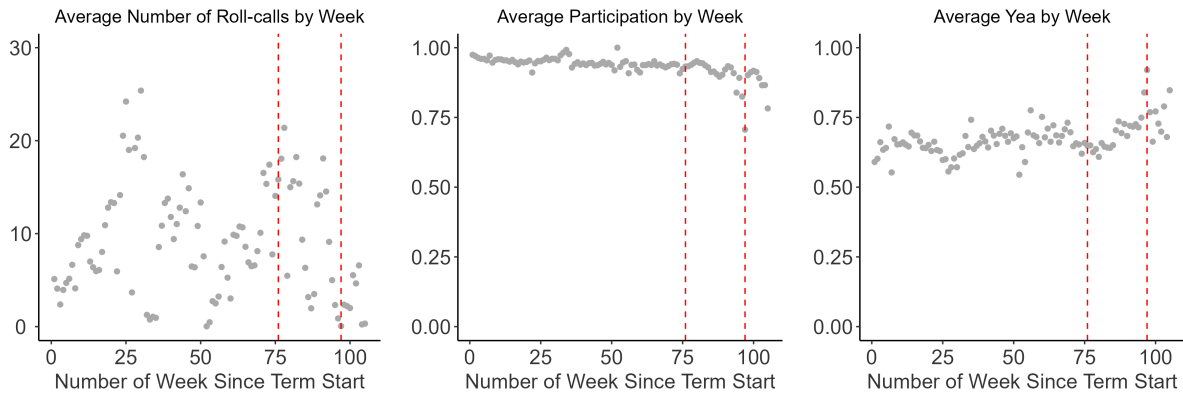
Figure A.3: Distribution of Topics of Bills Before (left) and After Primary (right)



term before (left) and after (right) primary elections. The topics are classified according to the Congressional Bills Project (Adler and Wilkerson, 2020). The distributions appear largely consistent across both periods, with government operations being the most frequently addressed topic, followed by defense, public lands, and other policy areas in a similar order. These patterns suggest that the legislative agenda remains relatively stable before and after primary elections.

Figure A.4 illustrates three aggregate characteristics of bills that reach a roll-call vote across a congressional term: the number of bills voted on per week (left), legislators' partici-

Figure A.4: Average Weekly Roll-call Voting Characteristics



Note: The x-axis of each figure represents the number of weeks since the start of a Congress. The first figure plots the average number of roll-call votes per week, the second displays the average participation rate, and the third shows the average yea vote rate. The two red vertical dotted lines indicate the time window between the average primary election date and the general election.

pation rates in those votes (center), and the frequency with which bills pass (right). The two red vertical lines indicate the average dates of the primary election and the general election, serving as reference points to assess whether the nature of bills considered changes before and after primary elections. While the plots reveal natural week-to-week fluctuations and a sharp shift near the general election, there is no clear evidence of a systematic or substantial change in bill characteristics around the primary election period.

A.10 Policy Position Results by Mean Differences Between Pre- and Post-Primary Periods

In the main article, I assess legislators’ changes in roll-call voting using a regression approach, estimating them through OLS coefficients. As a robustness check, I additionally measure changes in legislators’ voting behavior using a simpler method – calculating the mean difference in their voting patterns between the pre- and post-primary periods. This alternative approach ensures that the results are not driven by the specific estimation method used in the main analysis.

Specifically, for each legislator i in t , I measure a legislator’s change in voting pattern after primary election by the difference of average voting patterns in the pre- and post-primary periods. For example, if legislator i in t , participated 90% and 70% of roll-calls in the pre-primary period and post-primary period, respectively, then the change is 20% point decrease and I denote it as $Y_{it} = -0.2$.

In the second analysis, to ensure that the time windows are directly comparable, I restrict the sample to roll-calls occurring within the same specific time window in each year. For instance, if a primary election is held on August 1 and the general election on November 1, I compare the participation rate from August 1 to November 1 in the election year with the rate during that same period in the preceding year. As shown in Table A.7, the results remain largely consistent in both specifications.

Table A.7: Result on Policy Position by Mean Difference Before and After Primary Elections

Difference in Mean Before and After Primary					
	Regression Discontinuity				
	Change in Roll-call Participation	Change in Roll-call Extremism	Absolute Change in Roll-call Extremism	Change in Voting With Party Majority	Change in Voting With Party Leaders
Primary Win	0.127*** (0.018)	0.015 (0.012)	-0.021** (0.008)	0.014 (0.009)	0.013 (0.009)
Observations	926	918	918	920	913
Bandwidth	0.2	0.2	0.2	0.2	0.2
Specification	Linear	Linear	Linear	Linear	Linear
	Difference in Differences				
	Change in Roll-call Participation	Change in Roll-call Extremism	Absolute Change in Roll-call Extremism	Change in Voting With Party Majority	Change in Voting With Party Leaders
Primary Win	0.100*** (0.018)	0.010 (0.009)	-0.021*** (0.007)	0.013 (0.009)	0.008 (0.008)
Observations	3,714	3,694	3,694	3,698	3,638
Specification	Aggregated Legislator, Party-Cong.	Aggregated Legislator, Party-Cong.	Aggregated Legislator, Party-Cong.	Aggregated Legislator, Party-Cong.	Aggregated Legislator, Party-Cong.
Fixed Effects					
Difference in Mean Before and After Primary in Same Time Windows					
	Regression Discontinuity				
	Change in Roll-call Participation	Change in Roll-call Extremism	Absolute Change in Roll-call Extremism	Change in Voting With Party Majority	Change in Voting With Party Leaders
Primary Win	0.117*** (0.020)	0.008 (0.014)	-0.015 (0.010)	0.001 (0.010)	-0.007 (0.011)
Observations	920	913	913	915	908
Bandwidth	0.2	0.2	0.2	0.2	0.2
Specification	Linear	Linear	Linear	Linear	Linear
	Difference in Differences				
	Change in Roll-call Participation	Change in Roll-call Extremism	Absolute Change in Roll-call Extremism	Change in Voting With Party Majority	Change in Voting With Party Leaders
Primary Win	0.111*** (0.019)	0.009 (0.010)	-0.021*** (0.007)	0.006 (0.010)	-0.001 (0.010)
Observations	3,699	3,679	3,679	3,683	3,623
Specification	Aggregated Legislator, Party-Cong.	Aggregated Legislator, Party-Cong.	Aggregated Legislator, Party-Cong.	Aggregated Legislator, Party-Cong.	Aggregated Legislator, Party-Cong.
Fixed Effects					

* $p < 0.1$; ** $p < 0.05$; *** $p < 0.01$

Note: The table presents estimates of the difference in legislators' roll-call voting behavior before and after primary elections. The top panel compares all votes cast after the primary to those cast before. The bottom panel refines the comparison by examining votes cast post-primary against those from the same time window in the preceding year.

A.11 Aggregated Difference-in-Differences for Vote-Based Outcomes (Cross-Check)

The main-text difference-in-differences results for roll-call-based outcomes—participation, signed roll-call extremism, voting with the party majority, and voting with party leaders—are estimated at the legislator–bill vote level using equation (4) in the main text. As a cross-check, Table A.8 reports the corresponding estimates from an aggregated legislator–period specification identical in form to equation (5), with the dependent variable constructed in two steps: (i) for each legislator–Congress pair, a within-term regression of the vote-level outcome on an `AfterPrimary` indicator produces a legislator-specific coefficient; (ii) that coefficient is then used as the dependent variable in a DiD regression with `PrimaryWin` as the regressor, legislator fixed effects, and party–Congress fixed effects. This is the specification used in an earlier version of the manuscript and is retained here for readers who prefer the legislator-level summary.

Table A.8: Aggregated (β_{it} -Based) Difference-in-Differences for Vote-Based Outcomes

	Change in Roll-call Participation	Change in Roll-call Extremism	Change in Voting with Party Majority	Change in Voting with Party Leaders
Primary Win	0.100*** (0.018)	0.007 (0.008)	0.010 (0.008)	0.002 (0.008)
Observations	3,714	3,694	3,694	3,634
Winner Mean	0.028	−0.000	−0.000	0.001
Loser Mean	−0.091	0.007	−0.007	−0.003
Specification	Aggregated	Aggregated	Aggregated	Aggregated
Fixed Effects	Legislator, Party-Cong.	Legislator, Party-Cong.	Legislator, Party-Cong.	Legislator, Party-Cong.

* $p < 0.1$; ** $p < 0.05$; *** $p < 0.01$

Note: Each column reports the PrimaryWin coefficient from the aggregated specification in equation (5), applied to vote-based outcomes whose main-text estimates are reported at the vote level. Standard errors (in parentheses) are clustered by legislator. Unit of observation is the legislator–period; N refers to the number of legislator–Congress pairs for which the β_{it} outcome could be constructed. The aggregated participation estimate of 0.100 and the vote-level interaction of 0.115 in Table 3 are directly comparable: both identify the same within-legislator shift in post-primary participation between winners and losers. The same correspondence holds for the signed-extremism, voting-with-party, and voting-with-leader columns, where both specifications produce null estimates.

A.12 Non-Challenged Incumbents: Scope and Natural Placebo

Incumbents who face contested primaries may differ systematically from those who do not, which bears both on the scope of the main results and, by extension, on the lame-duck mechanism itself. I address this along two dimensions. First, I compare the two groups on the district-political characteristics most relevant to selection into a contested primary (general-election margin, district presidential vote) and on standard legislator covariates (Table A.9). Second, I extend the difference-in-differences specifications to a three-group categorical variable with primary losers as the reference and two non-loser dummies for narrow winners and non-challenged incumbents (Table A.10). The behavioral comparison provides a particularly sharp placebo: the lame-duck mechanism predicts that non-challenged incumbents—who never had a re-election threat to remove—should behave like narrow primary winners and unlike narrow primary losers. If they instead resembled losers post-primary, the mechanism would be in question.

The non-challenged group consists of incumbents who sought renomination without a contested primary, defined as those recorded as having a single-candidate primary or an unopposed primary (the latter recorded under a data convention that omits the runner-up entry, equivalent in substance), excluding Louisiana and Connecticut to match the main analysis ($N = 8,234$ legislator–Congresses, all with non-missing primary dates). For roll-call-based outcomes I apply the vote-level specification of equation (4) with $\text{Winner} \times \text{AfterPrimary}$ and $\text{NonChallenged} \times \text{AfterPrimary}$ interactions. For non-vote effort outcomes (sponsorship, cosponsorship, e-newsletters), pre- and post-primary counts for the non-challenged group are reconstructed from the same raw bill, cosponsorship, and *DCInbox* sources used in the main analysis, applying the identical symmetric window the original construction pipeline uses; sanity-check correlations against the existing main-analysis values exceed 0.9996 for all three outcomes.¹ The aggregated specifications follow equation (5) without the legislator fixed effects (consistent with the footnote there), which would otherwise absorb the Non-

¹For β_{it} -based outcomes (extremity, conservative voting, voting with party, voting with leader, abstention), within-Congress regression coefficients are already produced for both the contested and non-challenged samples by the existing pipeline, since the underlying fixed-effect regression operates on the full *MemberVotesCleaned* panel.

Table A.9: Scope Comparison on Observable Characteristics: Contested-Primary Incumbents vs. Non-Challenged Incumbents

	Group A: contested ($N = 3,860$)		Group B: non-challenged ($N = 8,234$)		
Variable	Mean	SD	Mean	SD	Difference
<i>Legislator characteristics</i>					
Age (years)	54.69	11.02	52.23	10.13	+2.47***
Tenure (prior Congresses)	4.71	4.36	4.01	3.87	+0.71***
Democrat (0/1)	0.622	0.485	0.515	0.500	+0.107***
DW-NOMINATE (1st dim.)	−0.062	0.400	0.002	0.378	−0.064***
DW-NOMINATE 1st dim. (extremity)	0.370	0.165	0.343	0.159	+0.027***
Redistricted (0/1)	0.246	0.431	0.209	0.406	+0.038***
<i>District political environment</i>					
Prior general-election vote share	0.698	0.148	0.688	0.149	+0.011***
District Dem. presidential vote share	0.520	0.154	0.494	0.134	+0.026***

* $p < 0.1$; ** $p < 0.05$; *** $p < 0.01$ (two-sided Welch t -test)

Note: Group A is the paper’s treatment sample (narrow primary winners and losers). Group B is the non-challenged sample defined as incumbents seeking renomination with a recorded primary date and either a single-candidate or an unopposed primary record (both indicate an uncontested primary; the latter arises from an unopposed single-candidate record that does not produce a runner-up entry). Louisiana and Connecticut are excluded in both groups to match the main analysis. The two district-environment variables (general-election margin, district presidential vote) are the central scope concern; their absolute differences across groups are within 1–2.6 percentage points.

Challenged dummy: every non-challenged incumbent is always coded as such, yielding no within-legislator variation on that indicator.

Table A.9 reports the comparison on observable characteristics. Statistically, contested-primary incumbents are modestly older, more senior, more likely to be Democrats, marginally more ideologically extreme, and more likely to have been redistricted, with all gaps reaching conventional significance under the very large combined sample. Substantively, the magnitudes are small: the largest gap is the Democrat share at +10.7 percentage points; the rest cluster around +0.03–+0.06 in standardized units. On the two district-political variables of central concern, the gaps are minimal. Prior general-election vote share differs by +1.1 percentage points (0.698 vs. 0.688) and district presidential Democratic vote share by +2.6 percentage points (0.520 vs. 0.494). The implicit concern that contested-primary incumbents might come from systematically more competitive electoral environments is not supported by the data; if anything, contested-primary incumbents represent slightly safer general-election seats.

Table A.10 reports the behavioral comparison across twelve outcomes. The pattern is consistent and theoretically expected. On every effort-related outcome—roll-call participation, abstention, the proportion of sponsoring, cosponsoring, and newsletter activity falling in the post-primary window—both narrow primary winners and non-challenged incumbents differ sharply from primary losers, and the magnitudes are similar across the two non-loser groups. On the directional ideology outcomes—voting with the party majority, voting with party leaders, change in conservative voting, and the signed extremity change—all three groups are statistically and substantively indistinguishable from zero. The lone exception is the absolute extremity change, where winners and non-challenged incumbents differ from losers by a small -0.013 and -0.012 : this gap is statistically detectable only under the large legislator–Congress sample, and winners and non-challenged remain indistinguishable from each other ($p = 0.136$). A formal test of equality between the Winner and NonChallenged interactions, $H_0 : \beta_W = \beta_{NC}$, fails to reject ($p > 0.10$) for nine of the twelve outcomes and is marginal ($p = 0.056$) for one (cosponsorship). The two rejections—e-newsletter activity and the signed extremity change—are small-magnitude differences in the same theoretical direction: both non-loser groups behave like primary winners rather than losers. Taken together with the scope comparison, the natural-placebo evidence supports the lame-duck interpretation: the post-primary effort decline is concentrated specifically among legislators whose re-election incentive disappears, not generated by some specific characteristic from contested primaries.

Table A.10: Three-Group Difference-in-Differences across Twelve Outcomes: Primary Losers, Winners, and Non-Challenged Incumbents

Outcome	Winner vs. Loser	Non-Challenged vs. Loser	Winner – Non-Challenged	<i>N</i>
<i>Panel A: Vote-level DiD (eq. 4). Coefficients on $Status \times AfterPrimary$.</i>				
Roll-call participation	0.114*** (0.014)	0.112*** (0.014)	0.002 (0.164)	11,209,104
Voting with party majority	−0.002 (0.004)	−0.002 (0.004)	0.000 (0.616)	10,295,321
Voting with party leader	−0.004 (0.004)	−0.004 (0.004)	0.000 (0.999)	8,970,645
<i>Panel B: Aggregated DiD on non-vote outcomes (eq. 5). DV is post-primary share of total activity.</i>				
Bill sponsorship	0.163*** (0.041)	0.168*** (0.040)	−0.005 (0.538)	6,976
Bill cosponsorship	0.097*** (0.017)	0.089*** (0.017)	0.008* (0.056)	4,759
E-newsletter	0.294*** (0.043)	0.317*** (0.043)	−0.023** (0.034)	1,869
<i>Panel C: Aggregated β_{it} DiD (eq. 5). DV is the within-Congress regression coefficient.</i>				
Change in roll-call abstention	−0.123*** (0.014)	−0.120*** (0.014)	−0.003 (0.109)	11,663
Change in conservative voting	−0.007 (0.006)	−0.005 (0.006)	−0.002 (0.127)	11,609
Change in extremity (signed)	−0.007 (0.006)	−0.009 (0.006)	0.002** (0.023)	11,609
Change in extremity	−0.013*** (0.005)	−0.012*** (0.005)	−0.001 (0.136)	11,609
Change in voting with party	0.007 (0.006)	0.006 (0.006)	0.001 (0.505)	11,609
Change in voting with party leader	0.005 (0.006)	0.003 (0.006)	0.002 (0.166)	11,444

* $p < 0.1$; ** $p < 0.05$; *** $p < 0.01$

Note: Each row reports estimates from a single regression with primary losers as the reference category. Coefficients are clustered at the legislator level; standard errors in parentheses. The third column reports the Winner – Non-Challenged difference (column 1 – column 2); the p -value in parentheses is from a cluster-robust Wald test of $\beta_W = \beta_{NC}$, and stars on the difference denote that test’s significance. Sample restricted to incumbents seeking renomination with a recorded primary date, excluding Louisiana and Connecticut (matching the main analysis). The non-challenged sample is defined as in Table A.9 ($N = 8,234$). Per the footnote to equation (5), the aggregated specifications drop legislator fixed effects, which would otherwise absorb the NonChallenged dummy. Cosponsorship is restricted to the Congresses with available source data (96–108).

A.13 Heterogeneity by Career Stage

A natural question is whether the within-Congress behavioral response to losing reelection incentives differs by legislator career stage—whether the effect of winning a close primary early in one’s career differs from the effect later in tenure. I construct seniority directly from the Voteview congressional member file by counting prior House terms for each legislator-Congress, splitting at *junior* (1st–2nd term, 0–1 prior terms) versus *senior* (3rd term or later, 2+ prior). I re-estimate both the main RDD specification and the simpler vote-level DiD on each subgroup, with a pooled fully-interacted model providing a formal differential test.

Table A.11 reports the results. In the RDD (Panel A, $|rv| < 0.2$), the participation effect is essentially identical across career stages: junior $\hat{\beta} = 0.134$ ($p < 0.001$, $N = 244$) versus senior $\hat{\beta} = 0.126$ ($p < 0.001$, $N = 682$), with a fully null differential ($p = 0.86$). None of the other seven outcomes display any meaningful differential by seniority, including the signed and absolute roll-call extremism measures. The vote-level DiD (Panel B), which trades some internal validity for substantially larger samples ($N \approx 3$ million vote-bill observations), agrees: the junior and senior participation responses are essentially the same ($\hat{\beta}_{\text{junior}} = 0.102$ vs. $\hat{\beta}_{\text{senior}} = 0.119$; differential $p = 0.44$), with no significant differential on any of the three ideology outcomes. Together the two specifications yield a clear answer: the within-Congress effect on participation is robust across career stages, and neither design detects a meaningful seniority differential. The close-primary sample is a binding power constraint for any sharper test of heterogeneity, as the wide differential confidence interval of the preferred RDD shows; but the much higher-powered vote-level DiD likewise finds no seniority differential, so the data rule out large qualitative differences across career stage.

Table A.11: Career-Stage Heterogeneity by Seniority

Outcome	Junior (1st–2nd term)	Senior (3rd term or later)	Diff. ($\beta_{\text{Win} \times \text{Sr}}$)	<i>N</i>
<i>Panel A: RDD ($rv < 0.2$, local-linear OLS, change/proportion outcome).</i>				
Roll-call participation	0.134*** (0.027)	0.126*** (0.023)	−0.008 (0.043)	926
Roll-call extremism	−0.020 (0.014)	0.011 (0.011)	0.031 (0.021)	918
Absolute extremity	−0.001 (0.010)	−0.009 (0.008)	−0.008 (0.015)	918
Voting with party	−0.005 (0.013)	0.005 (0.010)	0.011 (0.019)	918
Voting with leader	−0.010 (0.012)	0.004 (0.010)	0.014 (0.019)	911
Bill sponsorship	0.144 (0.166)	0.180** (0.087)	0.036 (0.181)	491
Bill cosponsorship	0.080 (0.089)	0.076* (0.046)	−0.004 (0.089)	287
E-newsletter	0.209 (0.128)	0.156 (0.095)	−0.053 (0.167)	210
<i>Panel B: Vote-level DiD (eq. (4), legislator + bill-by-party-by-Congress FE, cluster icpsr).</i>				
Roll-call participation	0.102*** (0.021)	0.119*** (0.016)	0.021 (0.027)	3,622,179
Roll-call extremism	−0.004 (0.005)	−0.005 (0.005)	0.002 (0.007)	3,296,336
Voting with party	0.002 (0.006)	−0.002 (0.005)	−0.004 (0.008)	3,290,805
Voting with leader	0.000 (0.005)	−0.006 (0.005)	−0.007 (0.008)	2,892,750

* $p < 0.1$; ** $p < 0.05$; *** $p < 0.01$

Note: Seniority constructed from the Voteview congressional member file as the count of prior House terms (same legislator) before the focal Congress. Standard errors in parentheses; Panel B is cluster-robust at the legislator level. The Diff. column reports the $\text{PrimaryWin} \times \text{Senior}$ coefficient from a fully-interacted model on the pooled sample. Subgroup *N*'s sum to less than pooled *N* because the differential model uses additional observations with non-missing outcomes that drop out of one or the other subgroup-only regression. Replication materials are provided with the paper.

A.14 Heterogeneity by Primary-to-General Window Length

A further question is whether the lame-duck effect persists beyond the short period between the primary and the general election. Persistence into a subsequent Congress is not directly estimable in my within-Congress identification: by construction, primary losers are not in $t+1$ as House members. A related and design-compatible question—whether the post-primary effect might be a brief blip rather than a sustained pattern—is whether the magnitude of the lame-duck response varies with the *length* of the post-primary window. State law sets primary dates anywhere from March to September, so close-primary losers face anywhere from about 8 weeks to about 8 months of lame-duck status before the general election.

I construct the number of days from each legislator’s primary date to the relevant November general-election day, restrict to plausible values (30–300 days; this drops two outlier rows), and split the close-primary RDD sample at the median of 147 days (≈ 5 months). I then re-estimate both the main RDD specification and the simpler vote-level DiD on each subgroup, with a pooled fully-interacted model providing a formal differential test. Table A.12 reports the results. The participation effect is statistically indistinguishable across the two window-length subgroups in both specifications: in the RDD, 0.119*** (late) versus 0.131*** (early), differential $p = 0.75$; in the vote-level DiD, 0.129*** (late) versus 0.106*** (early), differential $p = 0.29$. The two specifications give numerically opposing point estimates but neither differential is significant. The remaining outcomes are null or of negligible magnitude, including signed roll-call extremism in both panels. The one exception is absolute extremity: its early-window RDD cell is a small significant negative (-0.019 , $p < 0.05$), while the late-window cell and the timing differential are null (-0.021 , $p > 0.10$), indicating marginally smaller absolute ideological movement among long-window winners but no significant timing heterogeneity. The clearest pattern is e-newsletter activity, somewhat larger in the long-window subgroup (RDD: 0.298*** early vs. 0.073 late, differential $p = 0.11$; continuous-moderator interaction $p = 0.04$), consistent with the intuition that newsletter production accumulates naturally over a longer post-primary window. In the high-powered vote-level DiD, the with-leader early-window and differential coefficients are nominally significant (-0.012 and -0.022 , $p < 0.05$) but substantively trivial—one to two percentage

points—reflecting the very large vote-level sample rather than a meaningful positional shift. The substantive bottom line is that the lame-duck effect is largely *robust* to window length: a short window of weeks already delivers most of the response detectable in my data, and a longer window of several months does not produce a substantially larger one.

I caveat that primary date is set by state law and correlates with state political environment (Southern runoff states tend to hold later primaries, etc.), so the early-versus-late split is not orthogonal to underlying state characteristics. I re-estimate both panels including a $\text{South} \times \text{PrimaryWin}$ (and, for the DiD, $\times \text{AfterPrimary}$) interaction to absorb the dominant region confound; the differentials are essentially unchanged. Other state-level confounders may remain. The differential should be read as descriptive of timing heterogeneity rather than the causal effect of additional lame-duck time.

Table A.12: Heterogeneity by Primary-to-General Window Length

Outcome	Late primary (≤ 147 d to general)	Early primary (> 147 d)	Diff. ($\beta_{\text{Win} \times \text{Early}}$)	N
<i>Panel A: RDD ($rv < 0.2$, local-linear OLS, change/proportion outcome).</i>				
Roll-call participation	0.119*** (0.027)	0.131*** (0.025)	0.012 (0.037)	926
Roll-call extremism	0.015 (0.013)	-0.009 (0.012)	-0.025 (0.018)	918
Absolute extremity	0.002 (0.009)	-0.019** (0.009)	-0.021 (0.013)	918
Voting with party	0.003 (0.012)	0.004 (0.011)	0.001 (0.016)	918
Voting with leader	0.004 (0.012)	-0.003 (0.011)	-0.007 (0.016)	911
Bill sponsorship	0.267* (0.138)	0.098 (0.089)	-0.169 (0.158)	491
Bill cosponsorship	0.101 (0.070)	0.043 (0.047)	-0.058 (0.083)	287
E-newsletter	0.073 (0.095)	0.298*** (0.106)	0.225 (0.142)	210
<i>Panel B: Vote-level DiD (eq. (4), legislator + bill-by-party-by-Congress FE, cluster icpsr).</i>				
Roll-call participation	0.129*** (0.020)	0.106*** (0.018)	-0.029 (0.027)	3,622,179
Roll-call extremism	-0.002 (0.005)	-0.008 (0.006)	-0.005 (0.008)	3,296,336
Voting with party	0.007 (0.007)	-0.006 (0.005)	-0.012 (0.009)	3,290,805
Voting with leader	0.009 (0.007)	-0.012** (0.006)	-0.022** (0.010)	2,892,750

* $p < 0.1$; ** $p < 0.05$; *** $p < 0.01$

Note: Window length is the number of days from each legislator's primary date to the November general election (first Tuesday after the first Monday). Sample restricted to legislators with window length between 30 and 300 days (drops two implausible-date rows). Median split at 147 days (≈ 5 months). Standard errors in parentheses; Panel B is clustered at the legislator level. The Diff. column reports the $\text{PrimaryWin} \times \text{Early}$ coefficient from a fully-interacted model. Robustness controlling for $\text{South} \times \text{PrimaryWin}$ (Panel A) and $\text{South} \times \text{PrimaryWin} \times \text{AfterPrimary}$ (Panel B) leaves all differentials essentially unchanged. **Caveat:** primary date is endogenous to state political environment; the differential is descriptive, not causal. Replication materials are provided with the paper.

A.15 Days-in-Session Windows as Robustness

In measuring non-vote behavior, I used the symmetric pre/post-primary windows equalizing calendar days but not days the House is actually in session. Equalizing session days rather than calendar days is arguably the more appropriate comparison, and the empirical premise is real: averaged across the 3,860 contested-primary legislator-Congresses, my pre-primary windows contain a median of 64 House-active days while post-primary windows contain a median of 57 (median asymmetry +8 days; 69% of rows have shorter post windows; 24% exceed an absolute asymmetry of 20 days). If the estimates from the symmetric pre/post-primary design reflected fewer session days available for voting rather than reduced electoral incentive, equalizing session days should reduce the effect.

For each date from 1955 to 2024 ($n = 10,319$ dates), I construct a binary variable indicating whether the House was in active session on that date, combining Voteview House roll-call dates (a roll-call by definition occurs on a session day) with Library of Congress Congressional Record API queries filtered to records carrying a House Section. The resulting series matches the U.S. House Historian’s official Legislative Days within $\pm 5\%$ for all 68 Congress-Sessions in my analysis range (median absolute error 1.0 day, $r = 0.997$). For each (legislator, congress), I project the original symmetric calendar window onto this binary and trim each leg to $\min(\text{sd_pre}, \text{sd_post})$ session days. I then re-aggregate all six outcomes within the trimmed window and re-run the within-Congress RDD on the session-day-restricted measures. For the β_{it} -based ideology scores, I apply the same within-term score-construction procedure to the full Congress 84–117 roll-call panel restricted to the same per-legislator window, yielding session-day analogues of the change in conservative voting, in voting with the party majority, in voting with party leaders, and in participation. As a sanity check, with no window restriction the procedure reproduces the scores used elsewhere in the paper to machine precision over Congresses 84–92, the range for which a direct comparison is available.

Table A.13 reports the central comparisons. The participation effect is preserved and slightly amplified under session-day equalization across both legislator-level estimators. In

the within-Congress simple-mean RDD on $|rv| < 0.2$, the change-in-participation coefficient is 0.119*** at the calendar window and 0.125*** at the trimmed window; and in the β_{it} RDD on the participation outcome, the coefficient is 0.127*** at the calendar window and 0.132*** at the session-day window, again slightly amplified under session-day equalization. I do not subject the vote-level difference-in-differences to this windowing: it is estimated on individual roll-call votes rather than on a window-aggregated legislator-level outcome, so the session-day question (how many in-session days fall in each leg of the window) does not enter its construction, and the headline participation effect it records (Table 3) is a per-vote rate that is mechanically invariant to the number of session days. The ideology nulls central to the paper’s interpretation hold in the preferred specifications: across all window definitions the β_{it} change in signed roll-call extremism (own-pole), in voting with the party majority, and in voting with party leaders each satisfy $|\hat{\beta}| < 0.05$ and $p > 0.10$ (signed extremism 0.008 at the calendar window and 0.007 at the session-day window, both $p > 0.45$), consistent with the precise full-sample null (Table 4, with-party 0.003, with-leader 0.000). The Conservative-shift row reports the raw conservative-direction change in β_{it} , not signed by the legislator’s own party pole as the paper’s extremism measure is; its small negative coefficient is essentially identical at the two windows (-0.022 versus -0.024), so it too shows no session-day sensitivity, which is the quantity this robustness check evaluates.

One cell is sensitive to session-day equalization: simple-mean sponsorship RDD drops from 0.156** at the calendar window to 0.110 at the trimmed window ($p = 0.18$); cosponsorship and e-newsletter move in the opposite direction (cosponsorship stable at 0.10***; e-newsletter rising from 0.170** to 0.213***), so the broader pattern of increased non-vote activity is preserved. The absolute change in the β_{it} ideology score (an auxiliary measure of overall ideological movement distinct from the headline within-party and leader-following nulls) is a precise null at both windows (-0.008 at the calendar window, -0.003 at the session-day window; $p > 0.30$). I report both cells transparently; the headline participation increase and the headline ideology nulls hold across the windowing.

Table A.13: Pre/Post-Primary Outcomes under Calendar versus Session-Day-Equalized Windows

Outcome	Calendar window	Session-day window	Δ (sd – cal)	N
<i>Panel A: Simple-mean RDD ($rv < 0.2$, local-linear OLS, change/proportion outcome).</i>				
Roll-call participation	0.119*** (0.025)	0.125*** (0.027)	+0.006	940
Voting with party	0.021* (0.012)	0.008 (0.014)	–0.013	936
Voting with leader	0.019 (0.012)	0.014 (0.013)	–0.005	929
Bill sponsorship	0.156** (0.077)	0.110 (0.082)	–0.046	467
Bill cosponsorship	0.104*** (0.036)	0.097*** (0.036)	–0.007	431
E-newsletter	0.170** (0.072)	0.213*** (0.076)	+0.043	202
<i>Panel B: β_{it} RDD ($rv < 0.2$, OLS; Cong 84–117).</i>				
Conservative shift	–0.022** (0.011)	–0.024** (0.012)	–0.002	918
Signed extremism (β_{it} , own-pole)	0.008 (0.011)	0.007 (0.012)	–0.001	918
Absolute change ($ \beta_{it} $)	–0.008 (0.008)	–0.003 (0.009)	+0.005	918
With Party (β_{it})	0.017 (0.011)	0.011 (0.012)	–0.006	918
With Leader (β_{it})	0.015 (0.011)	0.017 (0.012)	+0.002	911
Participate (β_{it})	0.127*** (0.028)	0.132*** (0.029)	+0.005	940

Notes: Coefficient on PrimaryWin in both panels under each window definition. The calendar window runs from the pre-primary date equidistant to the general election through the general-election date; the session-day window is trimmed to the smaller of the pre- and post-primary counts of House-active days, centered on the primary date. Heteroskedastic OLS standard errors. Significance: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$. Replication materials are provided with the paper.

A.16 Placebo Decomposition of the Change Coefficient

As an additional check on the RDD identification, I conduct a placebo decomposition: separately estimating local-linear RDDs on the pre-primary and post-primary outcome *levels* at the same ± 0.2 bandwidth (Table A.14). The pre-primary RDD is a placebo: because the primary outcome is realized only after the primary date, narrow winners and losers should not differ in pre-primary behavior at the cutoff. For roll-call participation, the placebo passes cleanly: the pre-primary level coefficient is 0.02 ($p = 0.18$), while the post-primary level shows a large, significant jump (0.15***), so the headline change coefficient (0.129***) reflects a clean post-primary divergence. For the three count-based outcomes the placebo gives a more nuanced picture. Sponsorship passes (pre coef 0.19, $p = 0.85$). Cosponsorship and e-newsletters, however, show statistically significant negative pre-primary imbalances at the cutoff (-32.52 , $p = 0.009$ and -14.94 , $p = 0.048$, respectively) in the *opposite* direction of the headline effect: close-race winners cosponsor and circulate fewer items than losers in the pre-primary period, a pattern already visible descriptively in Table 2.

I treat this finding transparently. Three considerations bound its implication. First, the headline DV for these count-based activities is the post-primary share, $\text{post}/(\text{pre} + \text{post})$, rather than the raw count; the proportion is mechanically less sensitive to absolute-level imbalance. Second, the decomposition placebo is one of several identification checks, and the others are passed: the McCrary density test detects no manipulation of the running variable (Figure A.1); a placebo-window test confirms that the participation effect is a sharp outlier relative to estimates from randomly placed no-election windows ($p < 0.002$; subsection A.18); and the non-challenged three-group natural placebo confirms that legislators whose re-election incentive is *not* removed behave like primary winners, not losers, across twelve effort and ideology outcomes (Table A.10). Third, I directly verify each headline change coefficient by re-estimating the change RDD with the pre-primary level entered as a covariate on the same per-outcome sample (Table A.15). Six of the seven change coefficients remain robust: participation strengthens slightly ($0.129 \rightarrow 0.135^{***}$), sponsorship and e-newsletters are essentially unchanged ($0.163 \rightarrow 0.168^{**}$, $0.169 \rightarrow 0.167^{**}$), and the signed-extremism / with-party / with-leader nulls hold. Cosponsorship, originally only marginal

Table A.14: Decomposing the Change Coefficient: Pre-Primary and Post-Primary Levels

	Pre-Primary Level	Post-Primary Level	Change (post share)	N
Roll-call Participation	0.018 (0.013)	0.145*** (0.022)	0.129*** (0.019)	926
Sponsorship	0.19 (1.00)	0.31 (0.83)	0.163** (0.077)	491
Cosponsorship	-32.52*** (12.31)	-12.04 (11.99)	0.081* (0.042)	287
E-Newsletters	-14.94** (7.53)	6.55 (5.11)	0.169** (0.072)	210

* $p < 0.1$; ** $p < 0.05$; *** $p < 0.01$

Note: Each cell reports the local-linear OLS RDD coefficient on Primary Win at bandwidth ± 0.2 , with conventional standard errors in parentheses. The Pre-Primary Level and Post-Primary Level columns use the raw participation rate (Participation) or raw counts (Sponsorship, Cosponsorship, E-Newsletters) as the dependent variable; the Change column uses the post-primary share, post/(pre + post), identical to the Regression Discontinuity panel of Table 3.

at the headline ($p = 0.056$), drops to 0.066 ($p = 0.12$) and loses significance once pre-primary cosponsoring is partialled out, suggesting that this single estimate partly absorbs the absolute-level imbalance. I report the cosponsorship change transparently; the headline conclusions on participation, sponsorship, and e-newsletters continue to hold under this direct robustness check.

Applying the same decomposition to the three vote-level position outcomes returns null at both levels: for signed extremism, pre 0.014 ($p = 0.54$) and post 0.029 ($p = 0.24$); for with-party voting, pre 0.016 ($p = 0.33$) and post 0.024 ($p = 0.17$); for with-leader voting, pre 0.020 ($p = 0.22$) and post 0.032 ($p = 0.08$), all estimated at bandwidth ± 0.2 on the same legislator–Congress samples ($N = 918, 918$, and 911 respectively). The absolute-extremity change is an aggregated legislator–Congress measure with no vote-level pre/post-primary analog, so it is not decomposed here. Because both pre- and post-primary levels are small and at most marginally significant at the cutoff, the null on the headline change coefficient (RDD columns of Table 4) cannot be masking a constant winner–loser gap that is present

both before and after the primary and that differences away to zero in the change.

A.17 Robustness of Change Coefficients to Pre-Primary Level

In Table A.14 I documented that for the three count-based effort outcomes (sponsorship, cosponsorship, and e-newsletters), the pre-primary level RDD shows a negative imbalance at the cutoff: close-race winners are less active than losers in the pre-primary period. Because the headline outcome variable for these activities is the post-primary share, $\text{post}/(\text{pre}+\text{post})$, rather than the raw count, the change coefficient is mechanically less sensitive to absolute-level differences, but it is not fully immune. To verify directly, I re-estimate the change-outcome RDD adding the pre-primary level as an additional covariate, restricting both the baseline and the augmented specification to the same observations:

$$\Delta y_{ic} = \beta_0 + \beta_1 \text{PrimaryWin}_{ic} + \gamma_1 rv_{ic} + \gamma_2 \text{PrimaryWin}_{ic} \times rv_{ic} + \delta y_{ic}^{\text{pre}} + \varepsilon_{ic}. \quad (1)$$

If the change coefficient β_1 is largely unchanged when y_{ic}^{pre} is partialled out, the headline estimate is robust to the pre-imbalance. If β_1 shrinks substantially, the headline change is partly mechanical from the absolute-level imbalance.

The participation coefficient *strengthens* from 0.129 to 0.135 (both $p < 0.001$) when the pre-primary participation rate is partialled out; sponsorship (0.163 $\rightarrow$ 0.168) and e-newsletters (0.169 $\rightarrow$ 0.167) are essentially unchanged; the position outcomes (signed extremism, with-party, and with-leader) remain null in both specifications. Cosponsorship, however, drops from 0.081 ($p = 0.056$) to 0.066 ($p = 0.122$) and loses statistical significance once pre-primary cosponsoring is partialled out, suggesting that this single change estimate—originally only marginal—partly absorbs the pre-primary absolute-level imbalance. Overall, the headline conclusions on participation, sponsorship, and e-newsletters continue to hold under this direct robustness check; the cosponsorship finding should be interpreted with caution.

Table A.15: Change-Coefficient Robustness to Pre-Primary Level (Bandwidth ± 0.2)

Outcome	Coefficient on <i>Primary Win</i>		Pre-Level	N
	Baseline	+ Pre Control	Coefficient	
Roll-call Participation	0.129*** (0.019)	0.135*** (0.018)	-0.298*** (0.045)	926
Roll-call Extremism	0.003 (0.009)	0.004 (0.009)	-0.070*** (0.013)	918
With Party	0.003 (0.008)	0.004 (0.008)	-0.074*** (0.016)	918
With Leader	0.000 (0.008)	0.002 (0.008)	-0.057*** (0.016)	911
Sponsorship	0.163** (0.077)	0.168** (0.073)	-0.025*** (0.003)	491
Cosponsorship	0.081* (0.042)	0.066 (0.042)	-0.0004** (0.0002)	287
E-Newsletters	0.169** (0.072)	0.167** (0.073)	-0.0001 (0.0007)	210

* $p < 0.1$; ** $p < 0.05$; *** $p < 0.01$

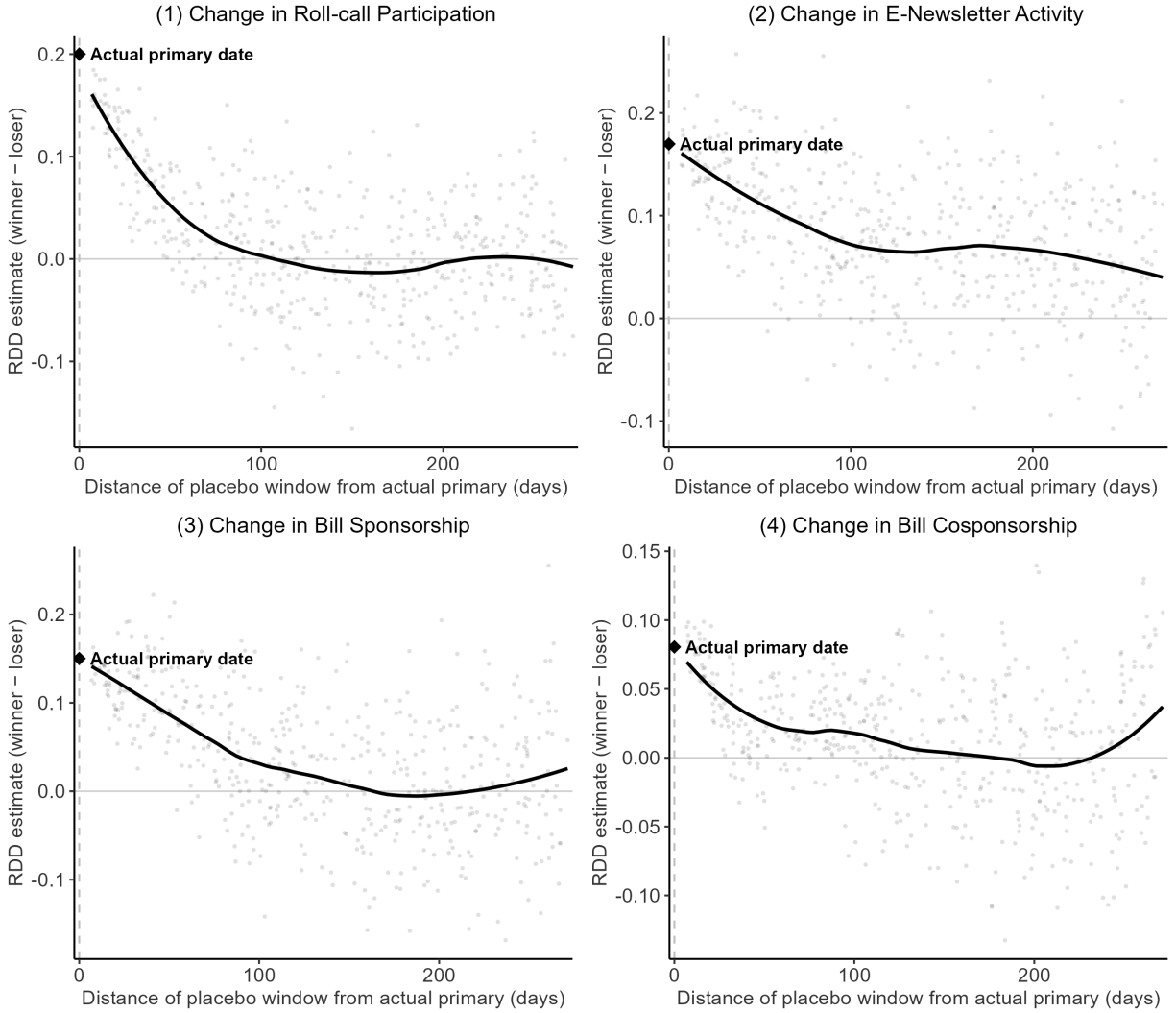
Note: Each row reports the coefficient on *Primary Win* from a local-linear OLS RDD at bandwidth ± 0.2 , with the change outcome as dependent variable. The Baseline column omits the pre-primary level term ($\delta = 0$ in equation (1)); the + Pre Control column includes it. Both columns are estimated on the same per-outcome restricted sample, so any difference reflects the pre-primary level partialling rather than sample composition. Conventional standard errors in parentheses; no clustering. Replication materials are provided with the paper.

A.18 Placebo Window Test

Because the pre- and post-primary windows are relatively short, the estimated effects could in principle reflect idiosyncratic variability in a particular short window rather than the primary date itself. To assess whether the estimated effects are specific to the actual primary date rather than an artifact of an arbitrary short window, I conduct a placebo-window test. For each placebo draw I displace every legislator’s primary date by a random amount toward the no-election portion of the term—holding each legislator’s own window length fixed—reconstruct the effort measures within the displaced window, and re-estimate the close-primary RD. If the effect is tied to the actual primary, windows placed well before it, when no election has yet occurred, should yield null estimates, and the actual-date estimate should be an outlier in the resulting distribution.

Figure A.5 plots, for each effort outcome, the RD estimate against the average distance between the placebo and actual primary dates. The pattern matches the identifying logic: estimates are largest when the window still straddles the actual primary and decay toward zero as it is moved into the no-election period, with the actual-date estimate (the diamond at distance zero) sitting at the top of each panel. Table A.16 quantifies the test against the null formed by placebo windows placed well before the primary (average displacement exceeding 150 days). The actual participation estimate is never matched by such a placebo window ($p < 0.002$), and sponsorship and e-newsletters are also outliers ($p = 0.045$ and $p = 0.031$). Cosponsorship is the exception ($p = 0.121$); this is the weakest and most pre-imbalanced effort outcome throughout the paper, and the result is consistent with the caveats already reported in subsection A.16. Because the placebo recomputes participation as a simple within-window rate change, the actual-date participation estimate shown here (0.20) corresponds to the simple-difference construction rather than the regression-adjusted headline (0.129***); both capture the same post-primary increase in effort.

Figure A.5: Placebo Window Test: RD Estimates Across Displaced Windows



Note: Each panel plots the close-primary RD estimate (winner – loser) for one effort outcome against the average distance, in days, between the placebo and actual primary dates. Gray points are individual placebo draws (random window displacements toward the no-election period); the solid line is a local-linear smooth. The diamond marks the actual-date estimate (distance zero).

Table A.16: Placebo Window Test: Actual-Date Estimate vs. the No-Election Null

Outcome	Actual estimate	Null mean	RI p -value
Change in Roll-call Participation	0.200	-0.004	< 0.002
Change in E-Newsletter Activity	0.170	0.061	0.031
Change in Bill Sponsorship	0.150	0.002	0.045
Change in Bill Cosponsorship	0.081	0.002	0.121

Note: The null distribution comprises placebo windows with an average displacement exceeding 150 days from the actual primary (windows placed in the no-election portion of the term). The RI p -value is the share of those placebo draws whose estimate is at least as large in absolute value as the actual-date estimate; $p < 0.002$ indicates that no placebo draw (of 500) matched the actual estimate. Participation uses the simple within-window rate change.

A.19 Previous Congress Placebo

As a further placebo, I repeat the analysis with the dependent variable from one Congress earlier. For each legislator at Congress t , I recompute the seven effort and position outcomes for the *previous* Congress $t - 1$ —a term in which that legislator had not yet faced the close primary I exploit—and re-estimate the within-Congress RD of the $t - 1$ outcome on the Congress- t treatment, winning or losing the primary. If the discontinuity I report were an artifact of fixed differences between narrow winners and narrow losers rather than the post-primary removal of the reelection incentive, it would appear at $t - 1$ as well. It does not. Table A.17 reports two samples: a sample that includes every legislator with a $t - 1$ record (Panel A), and the subset who also contested a primary in $t - 1$ (Panel B). The $t - 1$ outcomes are built from the same sources and design as the main analysis.

The coefficient is statistically insignificant for all seven outcomes in both samples (fourteen of fourteen null), confirming that the post-primary behavioral change is specific to the Congress in which the close primary occurs. Two patterns in the level columns are worth noting and are not placebo violations. First, the pre-primary participation level is marginally positive (0.022, $p \approx 0.10$), echoing the small baseline participation advantage that narrow winners hold over narrow losers even before the primary; the within-term *change* it implies is nonetheless zero. Second, the count-based pre-primary level for cosponsorship reproduces the same negative imbalance documented for the contemporaneous decomposition in subsection A.16, while sponsorship shows a modest positive pre-primary gap at $t - 1$ that has no contemporaneous counterpart. In both cases the imbalance is a difference in overall activity volume that the post-primary share absorbs, which is why the change column remains null.

Table A.17: Previous-Congress Placebo: Decomposed RD on Congress $t - 1$ Outcomes

Outcome	Pre-Primary Level	Post-Primary Level	Change (Headline)	N (Post)
<i>Panel A: Hybrid (all close-primary legislators with a $t - 1$ record)</i>				
Participation	0.022* (0.013)	0.020 (0.021)	-0.002 (0.018)	750
Signed Extremism	0.040 (0.025)	0.039 (0.026)	-0.001 (0.013)	747
With Party	-0.004 (0.018)	-0.008 (0.019)	-0.004 (0.009)	748
With Leader	0.002 (0.019)	0.006 (0.020)	0.006 (0.009)	739
Sponsorship	1.93* (1.09)	0.78 (0.79)	0.056 (0.083)	423
Co-sponsorship	-33.71** (14.86)	-22.31 (16.17)	0.045 (0.043)	232
E-Newsletter	7.89 (6.29)	7.72 (7.14)	-0.022 (0.099)	141
<i>Panel B: Contested-in-both (the $t - 1$ Congress also featured a contested primary)</i>				
Participation	0.033* (0.020)	0.042 (0.029)	0.009 (0.024)	365
Signed Extremism	0.044 (0.033)	0.045 (0.034)	0.001 (0.016)	363
With Party	0.000 (0.024)	0.001 (0.025)	0.001 (0.012)	364
With Leader	-0.001 (0.024)	0.012 (0.026)	0.013 (0.012)	362
Sponsorship	3.64** (1.58)	1.01 (0.92)	-0.057 (0.110)	212
Co-sponsorship	-43.99* (25.40)	-26.15 (26.54)	0.056 (0.061)	109
E-Newsletter	6.30 (8.25)	7.34 (7.42)	0.012 (0.120)	73

Note: For each close-primary legislator at Congress t , outcomes are measured at Congress $t - 1$ and regressed on the Congress- t treatment within $|rv| < 0.2$ (local-linear OLS, conventional standard errors in parentheses). Vote-based outcomes (participation, signed extremism, with-party, with-leader) use afterprimary-split means—participation over ≥ 10 available roll calls per side, positions over ≥ 10 cast votes—and the Change column is the post-minus-pre difference. Signed extremism is the own-party-pole indicator (higher = more own-pole voting); absolute extremity has no pre/post analog and is omitted, as in the contemporaneous decomposition (subsection A.16). Count-based outcomes (sponsorship, cosponsorship, e-newsletters) use the equidistant pre/post window: the Pre and Post columns are raw counts and the Change column is the post-primary share. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

A.20 Bootstrap Standard Errors for Headline Estimates

Several of my dependent variables are themselves estimated quantities — in particular, the within-term regression coefficients β_{it} that summarize a legislator’s ideology in a Congress are first-stage estimates that carry measurement error — so the analytical standard errors may understate the true uncertainty. As a check, I bootstrap the standard errors. I implement two bootstrap procedures at the same fixed bandwidth (± 0.2) used in the main RDD analysis, with $B = 1,000$ replications each:

- **Nonparametric bootstrap:** resample legislator–Congresses with replacement from the full close-primary sample and re-fit the local-linear RDD on the resampled data.
- **Wild bootstrap:** keep the original observations fixed and perturb fitted residuals by Rademacher weights (± 1 with probability 0.5 each), then re-fit the RDD on the perturbed dependent variable.

Table A.18 reports the resulting standard errors side by side with the analytical OLS SE and the corresponding two-sided p -values. For the four ideology outcomes the bootstrap SEs are roughly 50–70% larger than the analytical SEs, consistent with the concern that analytical OLS understates uncertainty when the dependent variable is itself estimated. None of the substantive conclusions changes: the participation effect remains highly significant ($p < 0.001$ under all three SEs); both ideology nulls (with-party, with-leader) remain null with bootstrap $p > 0.7$; and the auxiliary extremity nulls (signed and absolute) likewise remain null. For the three count-based outcomes, sponsorship is robust ($p \approx 0.03$ under both bootstraps); cosponsorship is, if anything, more clearly significant under the bootstraps ($p = 0.027$ nonparametric, $p = 0.019$ wild) than under the analytical SE ($p = 0.054$), so the bootstrap diagnostic does not corroborate the pre-imbalance sensitivity documented in subsection A.17 (the two checks ask different questions); e-newsletters move to the edge of conventional significance under the nonparametric bootstrap ($p = 0.058$) and remain significant at the 5% level under the wild bootstrap ($p = 0.040$).

Table A.18: Bootstrap Standard Errors at Bandwidth ± 0.2 ($B = 1,000$)

Outcome	Coefficient	Standard Error			Two-sided p -value		
		Analytical	NP Boot	Wild Boot	Analytical	NP Boot	Wild Boot
Roll-call Participation	0.129***	0.018	0.023	0.023	<0.001	<0.001	<0.001
Bill Sponsorship	0.163**	0.077	0.076	0.073	0.034	0.031	0.026
Bill Cosponsorship	0.081*	0.042	0.036	0.035	0.054	0.027	0.019
E-Newsletter	0.169**	0.072	0.089	0.082	0.019	0.058	0.040
Change in Extremity (signed)	0.003	0.009	0.014	0.013	0.742	0.834	0.827
Change in Extremity (abs.)	-0.008	0.006	0.010	0.011	0.195	0.418	0.442
Change in Voting With Party	0.003	0.008	0.013	0.013	0.715	0.824	0.818
Change in Voting With Leader	0.0004	0.008	0.013	0.013	0.952	0.970	0.969

* $p < 0.1$; ** $p < 0.05$; *** $p < 0.01$ (analytical SE)

Note: Each row reports the local-linear OLS RDD coefficient on Primary Win at fixed bandwidth ± 0.2 , with three SE estimates: (i) conventional analytical OLS, (ii) nonparametric bootstrap at $B = 1,000$ (resample legislator–Congresses with replacement and re-fit), (iii) wild bootstrap at $B = 1,000$ (Rademacher perturbation of residuals; observations held fixed). Two-sided p -values are computed from the z -statistic $\hat{\beta}/\text{SE}$ under the standard normal. Significance stars on the coefficient column are based on the analytical SE for consistency with Table 3 and Table 4. Replication materials are provided with the paper.

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